10 Urine Concentration and Dilution

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CHAPTER OUTLINE

KEY POINTS

- Nephron segments and vasculature in the renal medulla are arranged in complex but specific anatomic relationships, both in terms of which segment leads to the next segment and in terms of which segments are adjacent to one another. This plays an important role in the concentrating and diluting process.
- The urinary concentrating mechanism is dependent on two independent processes: 1. generation of a hypertonic medullary interstitium by concentration of NaCl and urea via countercurrent multiplication processes; and 2. osmotic equilibration of the tubule fluid within the medullary collecting ducts with the hypertonic medullary interstitium under the control of vasopressin.
- Vasopressin and the type 2 vasopressin receptor (V_2R) play a central role in the urinary concentrating mechanism. V_2R activation stimulates NaCl reabsorption by the thick ascending limbs of Henle, urea transport in terminal portions of the inner medullary collecting duct (IMCD), and accumulation of the water channel, AQP2, on the apical plasma membrane of collecting duct principal cells.
- Vasopressin binding to V_2R ultimately increases cytosolic cAMP levels and intracellular calcium. This stimulates AQP2 accumulation at the apical plasma membrane by inducing depolymerization of the actin cytoskeleton and by AQP2 protein phosphorylation.
- Vasopressin stimulates phosphorylation of the urea transporters UT-A1 and UT-A3 and their apical plasma membrane accumulation in the IMCD through two cAMP-dependent pathways: PKA and Epac (exchange protein activated by cAMP). This leads to increased urea permeability in the IMCD, which facilitates urea reabsorption, increasing medullary interstitial osmolality and the osmotic gradient promoting water reabsorption through AQP2.
- Urea is lost from the inner medullary interstitium, largely via the vasa recta, but urea recycling pathways play a major role in limiting this loss.
- Controversy persists as to the nature of the mechanism that generates the inner medullary osmolality, particularly the NaCl gradient, since there is no active NaCl transport in the thin ascending limb.

INDEPENDENT REGULATION OF WATER AND SALT EXCRETION

The independent regulation of water and solute excretion is essential for homeostatic functions of the kidney to be performed simultaneously. This means that in the absence of changes in solute intake or metabolic production of waste solutes, the kidney can excrete different volumes of water upon changes in water intake. This ability to excrete the appropriate amount of water without marked perturbations in solute excretion (without disturbing the other homeostatic functions of the kidney) is dependent on renal concentrating and diluting mechanisms, which form the basis of this chapter.

Kidney water excretion is tightly regulated by the peptide hormone arginine vasopressin (AVP). Under normal circumstances, the circulating AVP level is determined by osmoreceptors in the hypothalamus that trigger increases in AVP secretion when the osmolality of the blood rises above a threshold value, ~292 mOsm/kg H_2O (reviewed in Sands et al[1]). This mechanism can be modulated when other inputs to the hypothalamus (e.g., arterial underfilling, severe fatigue, and physical stress) override the osmotic mechanism. Upon an increase in plasma osmolality, AVP is secreted from the posterior pituitary gland into the peripheral plasma. The kidney responds to the variable AVP levels by altering urine flow. For example, during extreme antidiuresis (high AVP), water excretion is 100-fold lower than during water diuresis (low AVP). These changes are obtained without substantial changes in steady-state solute excretion (Fig. 10.1) and are made possible by the kidney's ability to concentrate and dilute the tubule fluid. During low circulating AVP levels, urine osmolality is less than that of plasma (290 mOsm/kg H_2O): the diluting function of the kidney. In contrast, when circulating AVP levels are high, urine osmolality is higher than that of plasma: the concentrating function of the kidney.

ORGANIZATION OF STRUCTURES IN THE KIDNEY RELEVANT TO URINARY CONCENTRATING AND DILUTING PROCESS

The kidney's ability to vary water excretion, without altering steady-state solute excretion, cannot simply be explained as a consequence of the sequential transport processes along the nephron.[2] The independent regulation of water and sodium excretion occurs in the renal medulla, where the nephron segments and vasculature (vasa recta) are arranged in specific anatomic relationships. The parallel interactions between nephron segments that occur as a result of its looped or hairpin structure need to be considered (Fig. 10.2).[3] In addition to structural arrangements, organized expression of water channels (aquaporins), urea transporters, and ion transporters along the renal tubule is important to urinary concentrating and diluting processes (Fig. 10.3). The role of many of these proteins in urinary concentrating and diluting mechanisms have been confirmed using genetically modified mice.[4] Detailed descriptions of the renal tubule system and vasculature are in Chapter 2; an overview of the anatomic relationships important for urinary concentration and dilution and some of their molecular features are provided here.

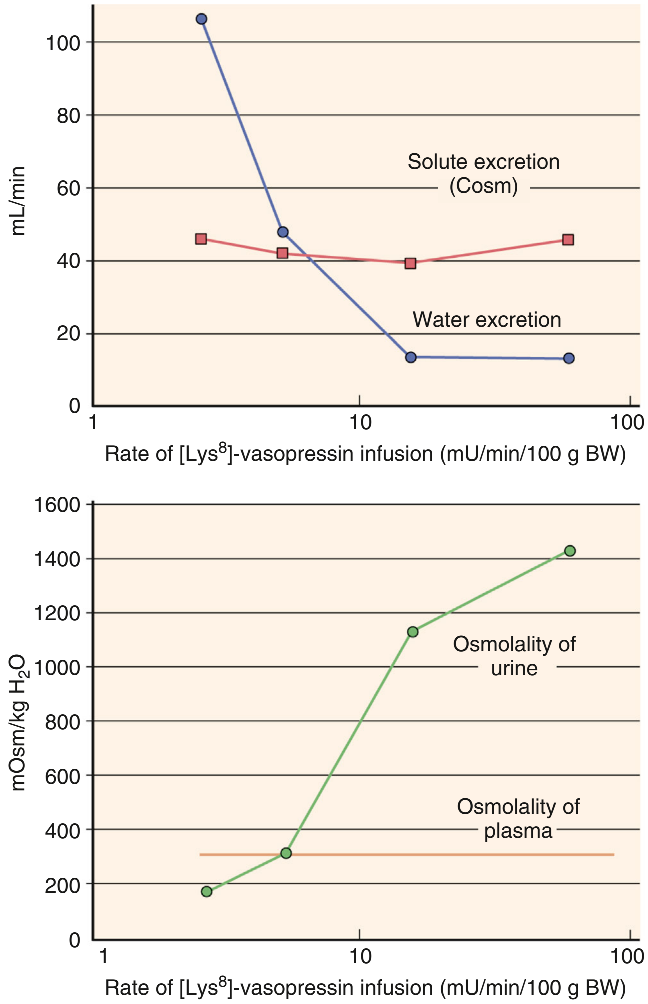

Fig. 10.1 Steady-state renal response to varying rates of vasopressin infusion in conscious rats. A water load (4% of body weight) was maintained throughout the experiments to suppress endogenous vasopressin secretion. Although the urine flow rate was markedly reduced at higher vasopressin infusion rates, the osmolar clearance (solute excretion) changed little. Concordantly, at higher vasopressin infusion rates, the osmolality of the urine increases significantly, whereas plasma osmolality remains constant. (Data from Atherton JC, Green R, Thomas S. Influence of lysine-vasopressin dosage on the time course of changes in renal tissue and urinary composition in the conscious rat. *J Physiol* 1971; 213(2):291–309.)

KIDNEY TUBULES

LOOPS OF HENLE

The kidney contains two populations of nephrons, long-looped and short-looped, which merge to form a common collecting duct system (see Fig. 10.2). Both types of nephrons have loops of Henle that are arranged in a hairpin configuration. Short-looped nephrons have glomeruli that are located superficially in the cortex and have loops that bend in the outer medulla. Long-looped nephrons have glomeruli that are located deeper within the cortex. Long-looped nephrons contain a

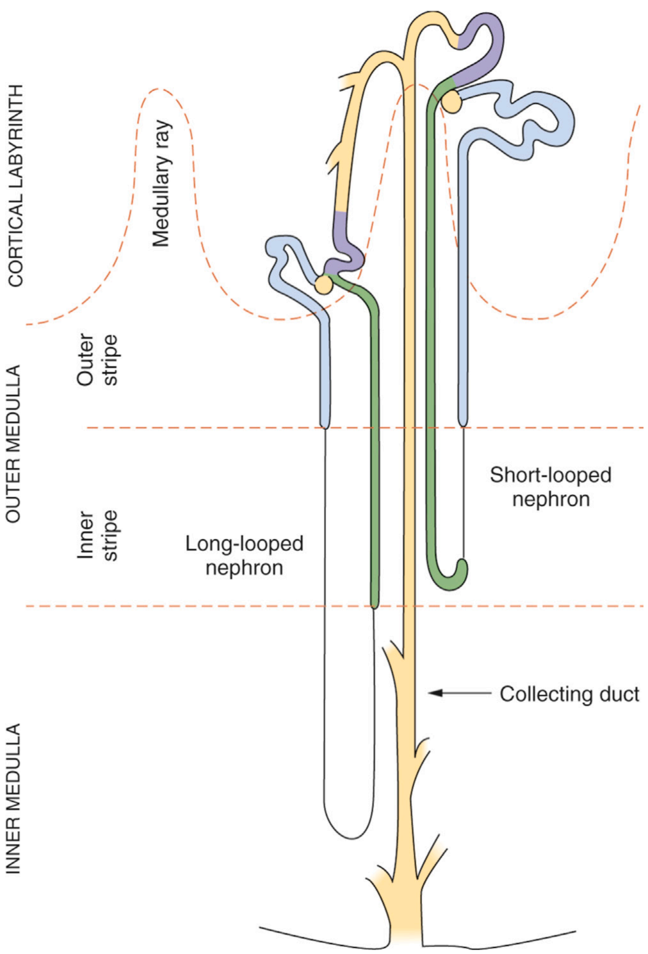

Fig. 10.2 Mammalian renal structure. Major regions of the kidney are shown on the left. Configurations of a long-looped and a short-looped nephron are depicted. The major portions of the nephron are proximal tubules (medium blue), thin limbs of loops of Henle (single line), thick ascending limbs of loops of Henle (green), distal convoluted tubules (lavender), and the collecting duct system (yellow). (Modified from Knepper MA, Stephenson JL. Urinary concentrating and diluting processes. In: Andreoli TE, Fanestil DD, Hoffman JF, Schultz SG. *Physiology of Membrane Disorders.* 2nd ed. New York: Plenum; 1986:713–726.)

thin ascending limb, found only in the inner medulla, whereas short-looped nephrons do not. The inner-outer medullary border is defined by the transition from thin to thick ascending limbs. Thus the outer medulla contains only thick ascending limbs, regardless of the type of loop. The long-looped nephrons bend at various levels of the inner medulla from the inner-outer medullary border to the papillary tip. Thus progressively fewer loops of Henle extend to deeper levels of the inner medulla. Some mammalian kidneys, including those in humans, also contain cortical nephrons whose loops of Henle do not reach into the medulla.

The loops of Henle receive tubular fluid from the proximal tubules. Tubular fluid exits the thick ascending limbs of both long- and short-looped nephrons (and from cortical nephrons) and flows into distal convoluted tubules. Thus the descending and ascending limbs of the loops of Henle have a countercurrent flow configuration (see Fig. 10.2). The descending thin limb of short-looped nephrons also differs structurally and functionally from the descending thin limb of long-looped nephrons.[5,6]

The location of the descending thin limb of short-looped nephrons within the outer medulla is illustrated in Fig. 10.4 (labeled in green).[7] The descending thin limbs of short-looped nephrons surround the vascular bundles in the outer medulla in a ringlike pattern (see Fig. 10.4 inset). Thin descending limbs of long-looped nephrons in the outer medulla differ morphologically and functionally from thin descending limbs of long-looped nephrons in the inner medulla.[8-11] The histologic transition from the outer medullary to the inner medullary type of thin descending limbs of long-looped nephrons is gradual and often occurs at some distance into the inner medulla, rather than strictly at the inner-outer medullary border as is the case for the transition between thin and thick ascending limbs. Various studies using immunohistochemical labeling and computer-assisted reconstruction have provided further details about the specialized functional architecture of loops in the inner medulla.[12-16] The use of various antibody-based markers in these studies has provided insights into the variable permeability of different thin limb segments to water and urea and how they may influence urine concentration and dilution.[17-21]

In addition to their unique structure and organization, a variety of transporters and channels in the loops of Henle are important for urine concentration and dilution. The kidney-specific chloride channel 1 (CLC-K1) localizes to both the apical and basolateral plasma membranes of thin ascending limbs,[22] where it mediates AVP-sensitive chloride conductance.[23] ClC-K1 null mice ($Clcnk1^{-/-}$) have greatly reduced transepithelial chloride transport in thin ascending limbs and produce large volumes of hypotonic urine even after water deprivation or AVP administration.[24] Inner medullary concentrations of Na^+ and Cl^- in $Clcnk1^{-/-}$ mice are approximately half those of controls. The findings in the $Clcnk1^{-/-}$ mice emphasize the importance of rapid chloride exit from thin ascending limbs in the inner medullary concentrating process and provide support for the "passive mechanism" (see later). The Na^+-K^+-$2Cl^-$ cotransporter type 2 (NKCC2) and the Na^+-H^+ exchanger type 3 (NHE3) are the major apical transporters mediating Na^+ entry in the thick ascending limb.[25-28] Total NHE3 knockout mice have a marked reduction in proximal tubule fluid absorption, where NHE3 is also expressed,[29-31] that manifests as a moderate increase in water intake associated with lower urinary osmolality. Renal tubule-specific NHE3 knockout mice or mice with selective deletion of NHE3 in the thick ascending limb have only small increases in fluid intake and urinary flow under basal conditions and a minor urinary concentrating defect.[32,33] NKCC2 is activated by AVP,[34] and in contrast to NHE3 knockout mice, NKCC2 knockout mice die before weaning due to renal fluid wasting and dehydration, highlighting the essential role of NKCC2 in the urinary concentrating mechanism.[35,36] The severe phenotype observed after deletion of NKCC2, relative to deletion of NHE3 that is responsible for reabsorption of far more Na^+, appears to be due to the role of NKCC2 in the mediation of tubuloglomerular feedback (TGF). An intact TGF allows NHE3 knockout mice to maintain a relatively normal distal fluid delivery through a decrease in glomerular filtration rate (GFR), while NKCC2 mice cannot compensate in this manner

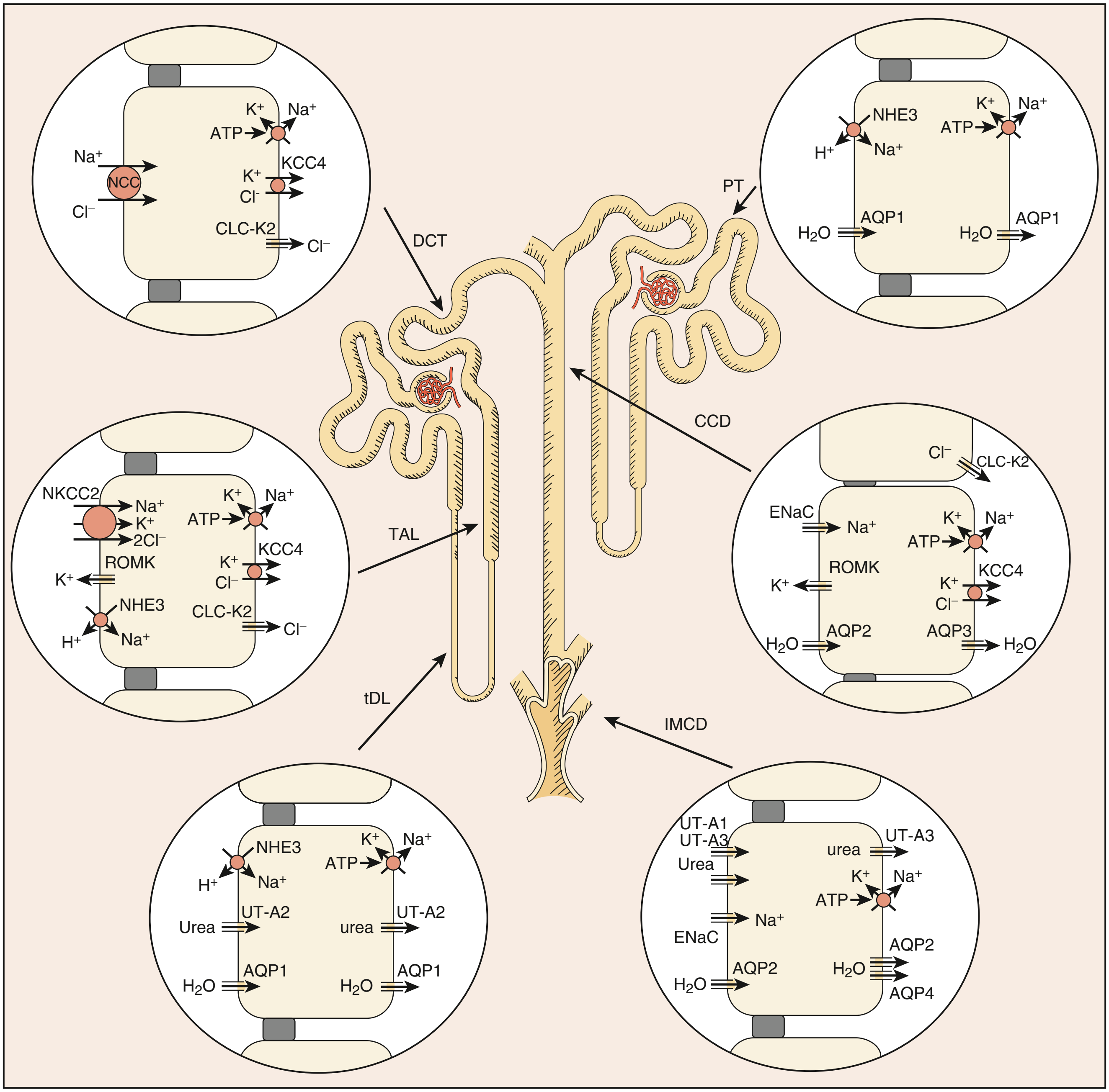

Fig. 10.3 Major aquaporins, urea transporters, and ion transporters/channels that are important to the urinary concentrating and diluting process. This is a schematic overview of a mammalian kidney tubule, showing the solute and water transport pathways in the proximal tubule (PT), thin descending limb of the Henle loop (tDL), thick ascending limb (TAL), distal convoluted tubule (DCT), cortical collecting duct (CCD), and inner medullary collecting duct (IMCD). The tubule lumen side is always on the left-hand side of the cell, whereas the interstitium is on the right-hand side. *Arrows* represent direction of movement. (Adapted from Fenton RA, Knepper MA. Mouse models and the urinary concentrating mechanism in the new millennium. *Physiol Rev.* 87:1083–1112, 2007.)

since the transporter is necessary for TGF to occur.[37,38] The renal outer medullary potassium channel (ROMK, Kir1.1) is an ATP-sensitive inwardly rectifier potassium channel that localizes to the apical plasma membrane of cells in the thick ascending limb, distal convoluted tubule, connecting tubule, and collecting duct system.[39-43] Chronic AVP treatment increases ROMK abundance in thick ascending limbs.[44,45] The majority of ROMK knockout mice die before weaning due to hydronephrosis and severe dehydration.[46] Those who survive to adulthood suffer from polydipsia, polyuria, impaired urinary concentrating ability, hypernatremia, and reduced blood pressure. This phenotype is consistent with the role of ROMK in active NaCl absorption in the thick ascending limb and countercurrent multiplication (see later).

DISTAL TUBULE SEGMENTS IN THE CORTICAL LABYRINTH

After fluid exits the loop of Henle, it enters the distal convoluted tubule within the cortical labyrinth. In most mammals, several distal tubules merge to form a connecting tubule arcade.[47] The connecting tubule cells express both the AVP-regulated water channel, aquaporin-2 (AQP2), and

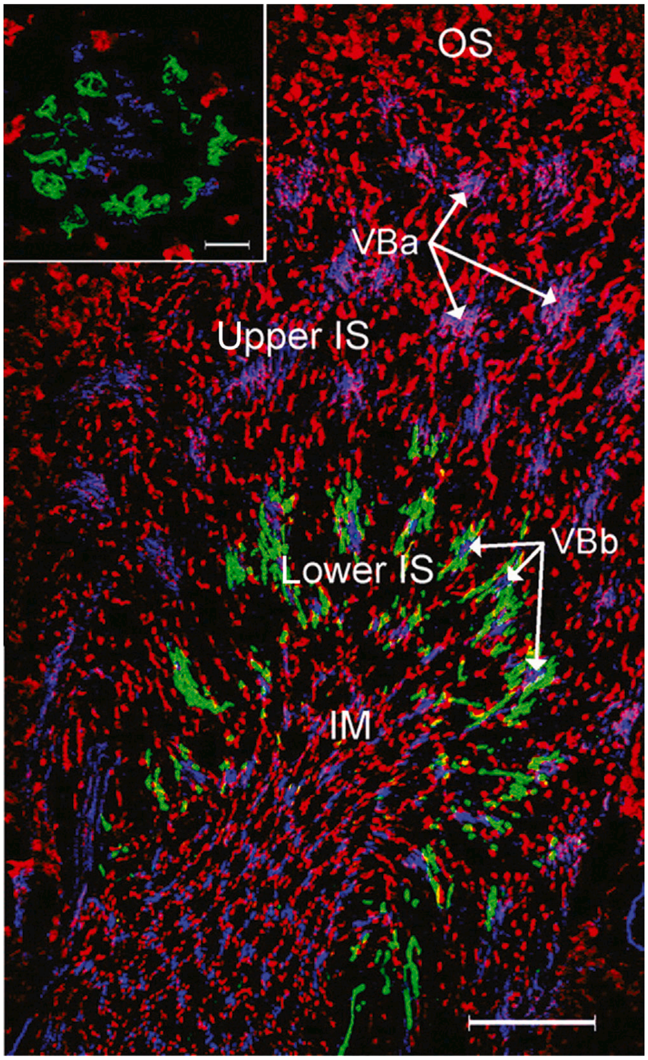

Fig. 10.4 Triple immunolabeling of rat renal medulla. Localization of UT-A2 (green), marking late thin descending limbs from short-looped nephrons, von Willebrand factor (blue) marking endothelial cells of vasa recta, and aquaporin-1 (red) marking thin descending limbs from outer medullary long-looped nephrons and early short-looped nephrons. Inset shows cross-section of vascular bundle demonstrating that UT-A2 positive thin descending limbs from short-looped nephrons surround vascular bundles in the deep part of the outer medulla. *IM,* Inner medulla; *IS,* inner stripe of outer medulla; *OS,* outer stripe of outer medulla; *VBa,* vascular bundles in outer part of inner stripe; *VBb,* vascular bundles in inner part of inner stripe. (From Wade JB, Lee AJ, Liu J, et al. UT-A2: a 55 kDa urea transporter protein in thin descending limb of Henle's loop whose abundance is regulated by vasopressin. *Am J Physiol Renal Physiol.* 2000;278:F52—F62.)

the type 2 vasopressin receptor (V_2R),[48] suggesting that they are sites of AVP-regulated water reabsorption. Tubular fluid exits the connecting tubules within the arcades and enters the collecting tubules within the superficial cortex and then the cortical collecting ducts. In many rodent species, several nephrons merge to form a single cortical collecting duct.[5,49]

COLLECTING DUCT SYSTEM

The collecting duct system, which starts in the cortex and runs to the tip of the inner medulla (see Fig. 10.2), is the major site of AVP-regulated water and urea transport and hence crucial to the urine-concentrating mechanism (see later). Collecting ducts are parallel to the loops of Henle in the medullary rays, outer medulla, and inner medulla. In general, the collecting ducts descend straight through the medullary rays and outer medulla without joining with other collecting ducts. In contrast, several collecting ducts merge as they descend within the inner medulla, resulting in a progressive reduction in the number of inner medullary collecting ducts from the inner-outer medullary border to the papillary tip.[49] The tapered structure of the renal papilla results from the reduction in collecting duct number, accompanied by a progressive reduction in the number of loops of Henle reaching the deepest levels of the inner medulla. In the rat, each inner medullary collecting duct is surrounded by approximately four ascending vasa recta.[50] One or two thin ascending limbs lie between each ascending vasa recta and opposite to the collecting duct.[50] Inner medullary collecting ducts in the inner medullary base form clusters that coalesce along the corticomedullary axis.[15,16,19,51-58] The thin descending limbs are predominantly present at the periphery of these clusters and appear to form an asymmetric ring around each collecting duct cluster, whereas the thin ascending limbs are distributed relatively uniformly among the collecting ducts and thin descending limbs.[55,59] Entrance and exit of descending and ascending thin limbs to collecting duct clusters appear to be important for the generation and maintenance of the osmolality gradient within the inner medulla.[18,50,60]

In addition to aquaporin water channels (see later), the collecting ducts express the epithelial sodium channel (ENaC).[61,62] AVP increases Na^+ reabsorption in the cortical collecting duct by increasing ENaC abundance, apical membrane expression, and open probability.[63-69] Mice with deletion of α ENaC from the collecting ducts alone have no difficulty in maintaining salt and fluid homeostasis,[70] whereas α ENaC deletion from the connecting tubule and collecting duct together results in a mouse model with increased urine volume and decreased urine osmolality.[71] This indicates that α ENaC activity within the connecting tubule is crucial for water homeostasis.

VASCULATURE

The major blood vessels that carry blood into and out of the renal medulla are named *vasa recta*. Blood enters the descending vasa recta from the efferent arterioles of juxtamedullary nephrons and supplies it to the capillary plexuses at each level of the medulla. The outer medulla capillary plexus is denser and better perfused than the plexus in the inner medulla.[72] Blood from the inner medullary capillary plexus feeds into the ascending vasa recta (which do not form directly from descending vasa recta in a looplike structure as depicted in many diagrams). Inner medulla ascending vasa recta traverse the inner stripe of the outer medulla in close physical association with the descending vasa recta in vascular bundles.[51] In many species, thin descending limbs of short-looped nephrons surround the vascular bundles[7] (see Fig. 10.4). The outer medullary capillary plexus is drained by vasa recta that ascend through the outer stripe of the outer medulla, separate from the descending vasa recta.[54] The arrangement of tubules and vessels in the vascular bundles is important for maintaining the interstitial osmolality gradient of the surrounding environment and urine concentration.[19,60,73]

The counterflow arrangement of the vasa recta in the medulla promotes countercurrent exchange of solutes and water, which is facilitated by the presence of aquaporin-1 (AQP1)[74,75] and UT-B urea transporters[58,76-79] in the endothelial cells of the descending portion of the vasa recta. Countercurrent exchange provides a means of reducing the effective blood flow to the medulla, important for the preservation of axial solute concentration gradients in the medullary tissue (see later) while maintaining a high absolute perfusion rate.[80] In contrast, the cortical labyrinth has a high blood flow. The vascular perfusion to this region promotes the rapid return of solutes and water reabsorbed from the nephron to the general circulation and helps maintain the interstitial concentrations of most solutes at levels close to those in the peripheral plasma. The medullary rays of the cortex have a capillary plexus that is considerably sparser than that of the cortical labyrinth, resulting in lower blood flow to the medullary rays.[2]

MEDULLARY INTERSTITIUM

The renal medullary interstitium connects the tubules and vasculature[59,81-83] and contains medullary interstitial cells, microfibrils, extracellular matrix, and fluid. The interstitium is relatively small in volume in the outer medulla and the outer portion of the inner medulla, which may help limit the diffusion of solutes upward along the medullary axis.[2,55,82] In contrast, the interstitial space is much larger in the inner half of the inner medulla[2,55,82] and contains large amounts of highly polymerized hyaluronic acid,[84] which may play a role in the generation of an inner medullary osmotic gradient.[84]

RENAL PELVIS

Urine exits the collecting duct system through the ducts of Bellini at the papillary tip and enters the renal pelvis (Fig. 10.5). The renal pelvis (or calyx in multipapillate kidneys) is a complex intrarenal urinary space that surrounds the papilla. The renal pelvis has portions that extend into the outer medulla, which are called fornices and secondary pouches. Although a transitional epithelium lines most of the pelvic space, the renal parenchyma is separated from the pelvic space by a simple cuboidal epithelium.[85] It has been proposed that water and solute transport could occur across this epithelium, thereby modifying the composition of the renal medullary interstitial fluid.[86] Two smooth muscle layers within the renal pelvic (calyceal) wall generate powerful peristaltic waves that appear to displace the renal papilla downward with a “milking” action[87] and may intermittently propel urine along the collecting ducts. The contractions compress all structures within the renal inner medulla including the interstitium, loops of Henle, vasa recta, and collecting ducts,[88] and may furnish part of the energy for concentrating solutes within the inner medulla, resulting in a concentrated urine[84] (see later).

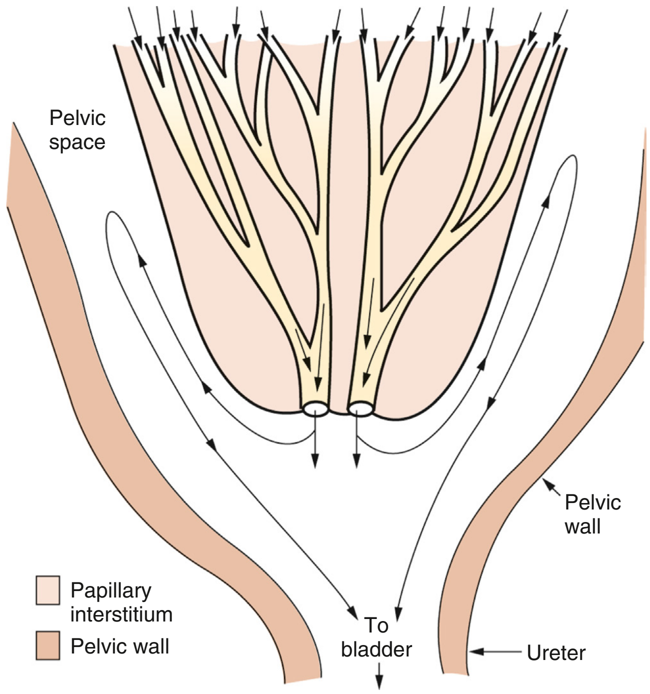

Fig. 10.5 Pattern of urine flow in papillary collecting ducts and renal pelvis. Urine exits the papillary collecting ducts (ducts of Bellini) at the tip of the renal papilla and is carried to the urinary bladder by the ureter. Under some circumstances, a fraction of the urine may reflux backward in the pelvic space and contact the outer surface of the renal papilla. Solute and water exchange across the papillary surface epithelium has been postulated (see text).

VASOPRESSIN AND THE TYPE 2 VASOPRESSIN RECEPTOR

The small peptide hormone AVP and the V_2R play a central role in the urinary concentrating mechanism. V_2R activation stimulates NaCl reabsorption by the thick ascending limbs of Henle, urea transport in terminal portions of the inner medullary collecting duct, and accumulation of AQP2 on the plasma membrane of collecting duct principal cells. These events permit the collecting duct luminal fluid to equilibrate osmotically with the surrounding interstitium in the kidney, resulting in water reabsorption and urine concentration. Dysfunction of this reabsorptive mechanism in the collecting duct results in the production of large amounts of dilute urine, about 15–20 L per day—a disease known as nephrogenic diabetes insipidus (NDI). Next, we address how V_2R and AQP2 interact via intracellular signaling pathways to regulate collecting duct water reabsorption and urine concentration (see “Clinical Relevance: Nephrogenic Diabetes Insipidus”).

ARGININE VASOPRESSIN

Secretion of AVP from the posterior pituitary is stimulated by an increase in plasma osmolality, but also by a reduction in plasma volume.[1] AVP activates regulatory systems necessary to retain water and restore osmolality to normal.[89] The effects of AVP occur through the stimulation of receptors on different cell types.[90,91] Here we focus on activation of the V_2R in renal epithelial cells for modulation of collecting duct water transport.

TYPE 2 VASOPRESSIN RECEPTOR

The V_2R is a seven transmembrane-spanning domain receptor that couples to heterotrimeric G proteins (GPCRs).[92,93] In the kidney it is expressed from the thick ascending limb cells of the loop of Henle to the collecting duct principal cells.[94-97] Expression of the V2R can be modulated epigenetically.[98] When AVP binds to the V_2R, adenylyl cyclase (AC) activity is stimulated and cytosolic cAMP levels increase.[99] Intracellular calcium is also increased by AVP via a mechanism involving calmodulin.[100-102] Together, these processes modulate accumulation of AQP2 in the apical plasma membrane of collecting duct principal cells, thus increasing transepithelial water permeability and facilitating osmotically driven water reabsorption (Fig. 10.6). A critical role of the V_2R for urinary concentration is confirmed by mouse models of X-linked nephrogenic diabetes insipidus (XNDI). Upon constitutive deletion,[103] male mutant mice display severe hypernatremia and significantly lower urine osmolality and die within 7 days after birth. Adult mice with conditional deletion of the V_2R[104,105] display all of the characteristic symptoms of XNDI[106,107] including polyuria, polydipsia, and resistance to the antidiuretic actions of AVP.

The function of the V_2R depends on interaction with heterotrimeric G proteins and β-arrestin. Upon AVP binding, the V_2R assumes an active configuration and the bound heterotrimeric G protein, Gs, dissociates into Gsα and Gsβγ subunits.[99] Cryoelectron microscopy studies have provided detailed insights into the AVP-V_2R-Gs signaling complex.[108,109] Various G proteins are localized to the basolateral plasma

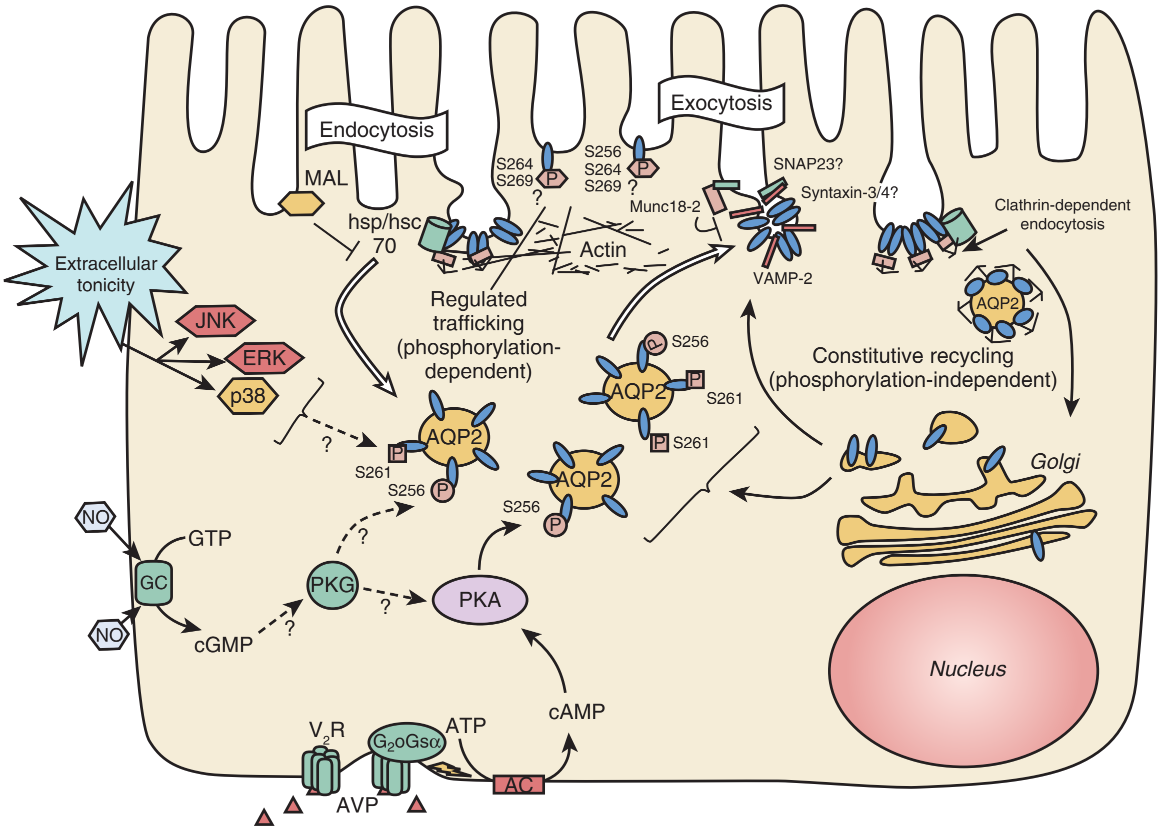

Fig. 10.6 Key events that contribute to the regulation of aquaporin-2 (AQP2) trafficking. The canonical pathway involves interaction of vasopressin (AVP) with the type 2 receptor (V_2R) on the basolateral surface of the principal cell. This increases cyclic adenosine monophosphate (cAMP) formation after $G_{\alpha s}$ stimulation of adenylyl cyclase (AC). Phosphorylation of AQP2 occurs initially on residue S256, via protein kinase A (PKA) activation. After vasopressin stimulation, residue S261 on AQP2 is dephosphorylated, and phosphorylation at S264 and S269 is increased. During exocytosis, AQP2 interacts with soluble *N*-ethylmaleimide–sensitive factor attachment protein receptor (SNARE) proteins and their regulatory proteins such as Munc18-2, and these interactions may be regulated by phosphorylation. At the cell surface, phosphorylated AQP2 is present in endocytosis-resistant domains and its interaction with heat shock protein/heat shock cognate 70 (hsp/hsc70), which is required for clathrin-mediated endocytosis, is inhibited. The myeloid and lymphocyte protein (MAL) is also involved in AQP2 endocytosis by an as-yet-unknown mechanism. Constitutive exocytosis of AQP2 occurs without vasopressin stimulation and does not require AQP2 phosphorylation on residue S256. Accumulation of AQP2 at the plasma membrane is increased by inhibiting clathrin-mediated endocytosis. AQP2 phosphorylation can also be increased by stimulating the cyclic guanosine monophosphate/protein kinase G (cGMP/PKG) pathways using, for example, nitric oxide (NO). Extracellular hypertonicity activates the mitogen-activated protein (MAP) kinase pathway, and c-Jun N-terminal kinase (JNK), extracellular signal–regulated kinase (ERK), and p38 MAP kinase activities are all required for AQP2 surface accumulation after acute hypertonic shock. Finally, AQP2 trafficking involves the actin cytoskeleton and actin depolymerization results in cell-surface accumulation of AQP2 without the need for vasopressin stimulation. *ATP,* Adenosine triphosphate; *GC,* guanylyl cyclase; *GTP,* guanosine triphosphate; *SNAP23,* synaptosomal-associated protein 23; *VAMP-2,* vesicle-associated membrane protein 2.

membrane of the thick ascending limb of Henle, distal convoluted tubule, and collecting duct principal cells.[110,111] Activated Gsα stimulates adenylyl cyclase 6 (AC-6) and cAMP levels are increased.[112] Mice lacking AC-6 have significant water balance abnormalities.[113,114] After AVP binding,[115] various accessory proteins aid in V_2R internalization and degradation, thus terminating the response.[99,116-121] Destruction of cAMP by cytosolic phosphodiesterases is also associated with limiting V_2R responses,[122] but the V_2R can continue to signal from endosomes after internalization.[120]

A critical step in V_2R internalization is the binding of β-arrestin to the V_2R,[123] which is triggered by phosphorylation of V_2R by various kinases.[124] Following β-arrestin–dependent ubiquitylation of the V_2R,[125] arrestin-receptor complexes recruit the clathrin adaptor protein AP-2[117] and the complex is then internalized via clathrin-mediated endocytosis.[118,126,127] Arrestins also uncouple GPCRs from heterotrimeric G proteins, producing a desensitized receptor.[128] Restoration of prestimulation levels of V_2R at the cell surface requires several hours.[129-131] The majority of the V_2R that is internalized with AVP is degraded,[125,132,133] and delivery of both AVP and the V2R is required to terminate the signaling response.[134]

Clinical Relevance: Nephrogenic Diabetes Insipidus

Nephrogenic diabetes insipidus (NDI) results from the inability of the kidney to respond to AVP and produce concentrated urine. Congenital NDI results from mutations in the V_2R in 90% of families and in AQP2 the others.[137] Acquired forms of NDI occur much more frequently and arise as a consequence of drug treatments, electrolyte disturbances, and urinary tract obstruction. In most manifestations of acquired NDI, dysregulation of AQP2, in terms of either protein abundance or AQP2 membrane targeting, plays a fundamental role in the development of polyuria.[136] Reduction of AQP2 in acquired NDI is most likely the primary cause of the NDI, rather than being a secondary event (e.g., as a consequence of the increased urine production or reduction in interstitial osmolality). For example, in models of hypokalemic and lithium-induced NDI, the changes in AQP2 expression in the kidney cortex are identical to those seen in the inner medulla,[484-486] which indicates that interstitial tonicity is not a major factor. Moreover, washout of the medullary osmotic gradient for 1 or 5 days using the loop diuretic furosemide has no effect on AQP2 expression,[486,487] which indicates that high urine flow in itself is not responsible for the reduced AQP2 expression in experimental NDI. Studies investigating the molecular physiology and signaling pathways regulating water and urea transport have identified several novel therapeutic possibilities for treating NDI.[136,427]

VASOPRESSIN-REGULATED WATER TRANSPORT

COLLECTING DUCT WATER ABSORPTION AND OSMOTIC EQUILIBRATION

Urine concentration depends on 1. generation of a hypertonic medullary interstitium by concentration of NaCl and urea via countercurrent processes; and 2. osmotic equilibration of the tubule fluid within the collecting ducts, first with the isotonic cortical interstitium followed by equilibration with the hypertonic medullary interstitium. When circulating AVP levels are low, the water permeability of the collecting ducts is low; as a result, relatively little water is reabsorbed from the tubule fluid and large volumes of hypotonic urine are produced. In contrast, high circulating AVP levels increase the permeability of the apical membrane of the thick ascending limb to NaCl, leading to an increase in the osmolality of the peritubular interstitium (due to countercurrent multiplication). AVP also increases the water permeability of the collecting ducts to high levels. Together, this results in water being reabsorbed from the cortical and outer medullary portions of the collecting duct system via aquaporin water channels,[135-137] resulting in the production of a small volume of hypertonic urine, with osmolality approaching that of the inner medullary interstitium.

The late distal tubule (late distal convoluted tubule, connecting tubule, and initial collecting tubule) is the earliest site along the renal tubule where water absorption increases during antidiuresis (Fig. 10.7).[138] The distal convoluted tubule does not express any water channels, but it does express the V_2R and AVP regulates NaCl transport in this segment via increasing activity of the NaCl cotransporter, NCC.[139,140] In contrast, the connecting tubule and cortical collecting duct express the V_2R and AQP2.[94] Thus these segments are likely the earliest sites of distal tubular osmotic equilibration.

The volume of water absorption in the connecting segment and initial collecting tubule required to raise tubule fluid to isotonicity is considerably greater than the additional amount required to concentrate the urine above the osmolality of plasma in the medullary portion of the collecting duct system.[2] Consequently, during antidiuresis, most of the water reabsorbed from the collecting duct system enters the cortical labyrinth, where the effective blood flow is high enough to return the reabsorbed water to the general circulation without diluting the interstitium. In contrast, if such a large volume of water was reabsorbed along the medullary collecting ducts, there would be a significant dilution effect on the medullary interstitium, thereby impairing the concentrating ability.[141,142]

During water diuresis, a modest corticomedullary osmolality gradient persists[143,144] and the water permeability of the collecting ducts is low but not zero.[145,146] Consequently, some water is reabsorbed by the collecting ducts during water diuresis. Most of this water reabsorption occurs in the terminal inner medullary collecting ducts, where the transepithelial osmolality gradient is highest. In fact, it is almost paradoxical that more water is absorbed from the terminal inner medullary collecting ducts during water diuresis than during antidiuresis, owing to a much greater transepithelial osmolality gradient.[141,142,147]

AQUAPORIN WATER CHANNELS

AQUAPORIN-1

AQP1 was identified in 1991 by Peter Agre and associates.[148-151] AQP1 is expressed in proximal tubules, thin descending limbs of long loop nephrons[9,152,153] but not short loop nephrons,[17] and descending vasa recta.[74] Proximal tubule fluid absorption is markedly reduced in AQP1 knockout mice, but distal

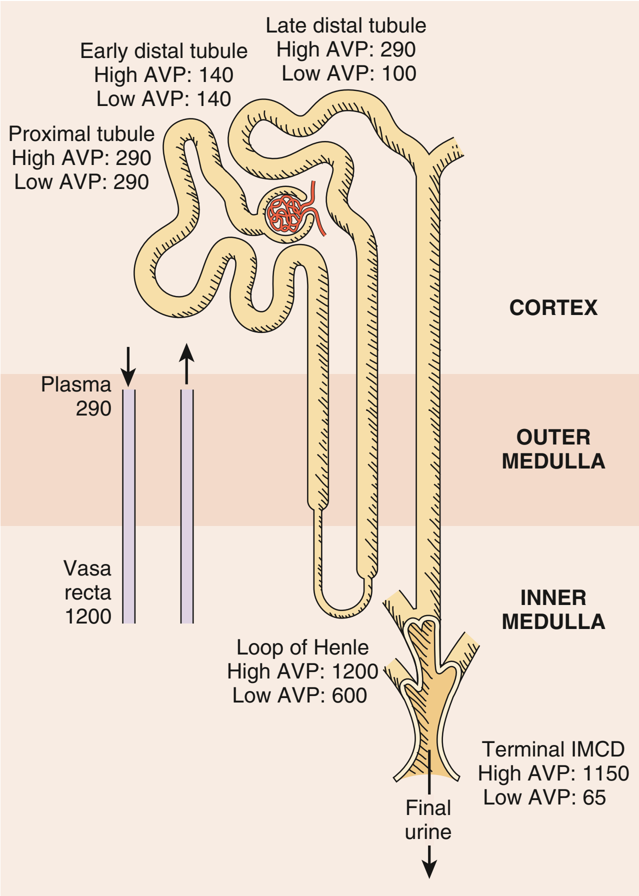

Fig. 10.7 Typical osmolalities (in mOsm/kg H_2O) found in various vascular (left) and renal tubule (right) sites in rat kidneys. Fluid in the proximal tubule is always isosmotic with plasma (290 mOsm/kg H_2O). Fluid emerging from the loop of Henle (entering the early distal tubule) is always hypotonic. Osmolality in the late distal tubule increases to plasma level only during antidiuresis. Final urine is hypertonic when the circulating vasopressin level is high and hypotonic when the vasopressin level is low. A high osmolality is always maintained in the loop of Henle and vasa recta. During antidiuresis, osmolalities in all inner medullary structures are nearly equal. Osmolalities are somewhat attenuated in the loop and vasa recta during water diuresis (not shown). *AVP*, Vasopressin; *IMCD*, inner medullary collecting duct.

delivery of water and NaCl is not increased due to a reduction in GFR via the tubular-glomerular feedback mechanism.[154] The osmotic water permeability of isolated perfused thin descending limbs from AQP1 knockoutmice is also reduced compared with controls.[155] Descending vasa recta also display a marked reduction in osmotic water permeability in AQP1 knockout mice.[74,75] Hence both countercurrent multiplication and countercurrent exchange processes are likely to be impaired in AQP1 knockout mice, consistent with their urinary concentrating defect phenotype that does not respond to water deprivation.[156] Consistent with this, humans with a loss-of-function mutation in AQP1 are unable to maximally concentrate their urine when challenged with water deprivation, although they have no obvious clinical phenotype under normal conditions.[157]

AQUAPORIN-2

AQP2 was cloned in 1993 and localized to collecting duct principal cells[158-160] (Fig. 10.8). Stimulation of the collecting duct with AVP results in the accumulation of AQP2 on the plasma membrane of principal cells (Fig. 10.9), in a process involving the recycling of AQP2 between intracellular vesicles and the cell surface[135,161-167] (see later). AQP2 localizes to the basolateral plasma membrane in some regions of the collecting duct[159,168-172] (see Fig. 10.8), where its expression is increased by AVP[173,174] or chronic administration of aldosterone.[171,175,176] Basolateral AQP2 is a potential water transport pathway across the basolateral membrane, but a proportion also likely represents a transient step in an indirect apical targeting pathway for the AQP2 protein (Fig. 10.10).[177,178] In addition, basolateral AQP2 may have a role in cell migration and tubulogenesis.[179,180]

Mutations in the *AQP2* gene are the cause of autosomal-dominant or autosomal-recessive NDI.[181] A number of different genetic models have been generated to assess the role of AQP2 in the urinary concentrating mechanism, including inducible and nephron-specific models of AQP2 deletion, models where essential phosphorylation sites in AQP2 are modified, and models of autosomal-dominant and autosomal-recessive NDI.[182-191] The major phenotype in these models is severe polyuria; however, with free access to water, plasma concentrations of electrolytes, urea, and creatinine are not different in knockout mice compared with controls. In an alternative mouse model with connecting tubule-specific AQP2 deletion, the role of the connecting tubule in regulating body water balance under basal conditions, but not for maximal concentration of the urine during antidiuresis, was uncovered.[192] Taken together, several mouse models confirm that AQP2 is responsible for most of the transcellular water reabsorption in the connecting tubule and collecting duct system.

AQUAPORIN-3 AND AQUAPORIN-4

Water absorbed via AQP2 leaves the cells via the basolateral water channels AQP3 and AQP4, which render the basolateral membrane of principal cells constitutively permeable to water.[193-196] AQP3 expression is predominantly in the cortex and decreases toward the inner medulla, whereas a reverse pattern is seen in AQP4, the abundance of which is highest in the inner medulla[194,196] (see Fig. 10.8). The abundances of AQP3 and AQP4 can be increased by the long-term action of AVP or dehydration.[193,196-199]

The osmotic water permeability of the basolateral membrane of cortical collecting ducts isolated from AQP3 knockout mice is reduced compared with control mice.[200] Consequently, AQP3 knockout mice are markedly polyuric.[2] AQP4 knockout mice have a fourfold decrease in inner medullary collecting duct osmotic water permeability relative to controls, indicating that AQP4 is responsible for the majority of water movement across the basolateral membrane in this segment.[201,202] Nonetheless, AQP4 knockout mice have no basal difference in urine osmolality. However, after 36 hours of water deprivation, AQP4 knockout mice have a significantly reduced maximal urine osmolality that cannot be further increased by AVP administration. This modest decrease in urinary concentrating ability in AQP4 knockout mice, compared with the profound concentrating defect in AQP3 knockout mice, is likely due to the normal distribution of water transport along the collecting duct.[2] Much greater osmotic reabsorption of water occurs in the cortical portion of the collecting duct system (where

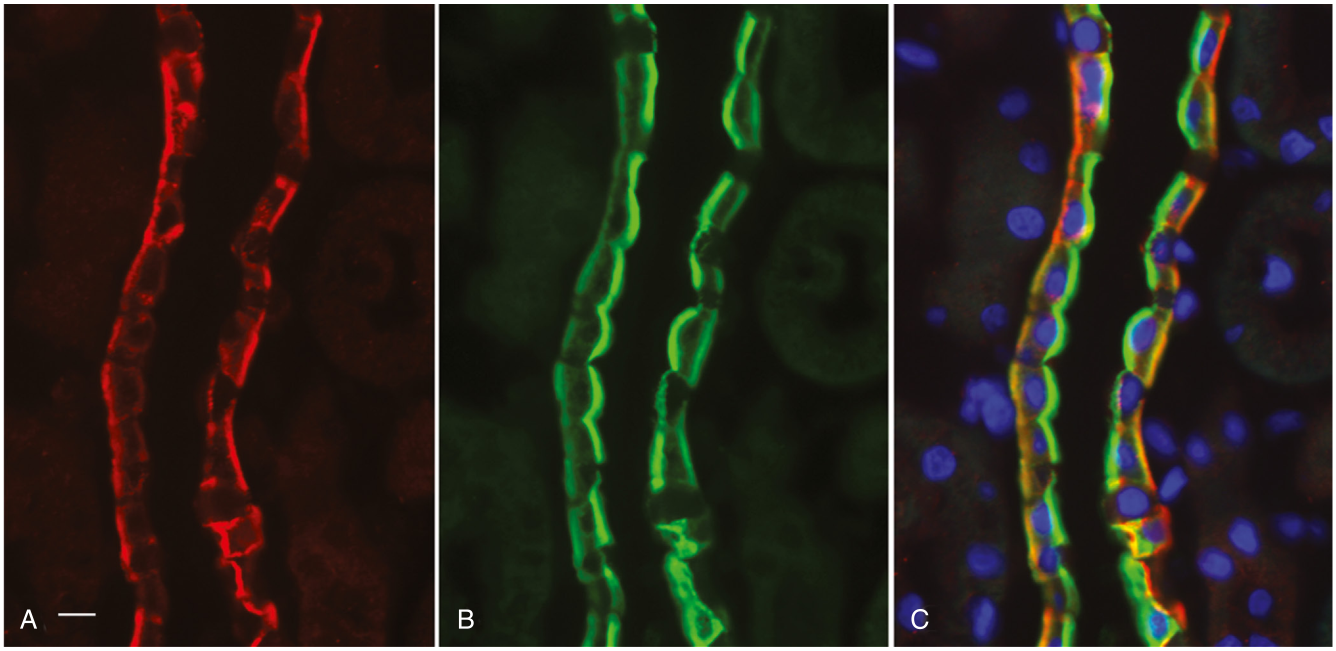

Fig. 10.8 Localization of aquaporins in the outer medullary collecting duct (outer stripe) of rat kidney. Images show sections immunostained in (A**)** for aquaporin-4 (AQP4) *(red)* and (B) for aquaporin-2 (AQP2) *(green).* The merged image in (C) shows that AQP2 is largely apical in this region, but both AQP2 and AQP4 are present on basolateral membranes. Intercalated cells are not stained with either antibody and appear as darker gaps among the other cells. (C) nuclei are stained with 4′,6-diamidino-2-phenylindole (DAPI). Bar equals 10 μm.

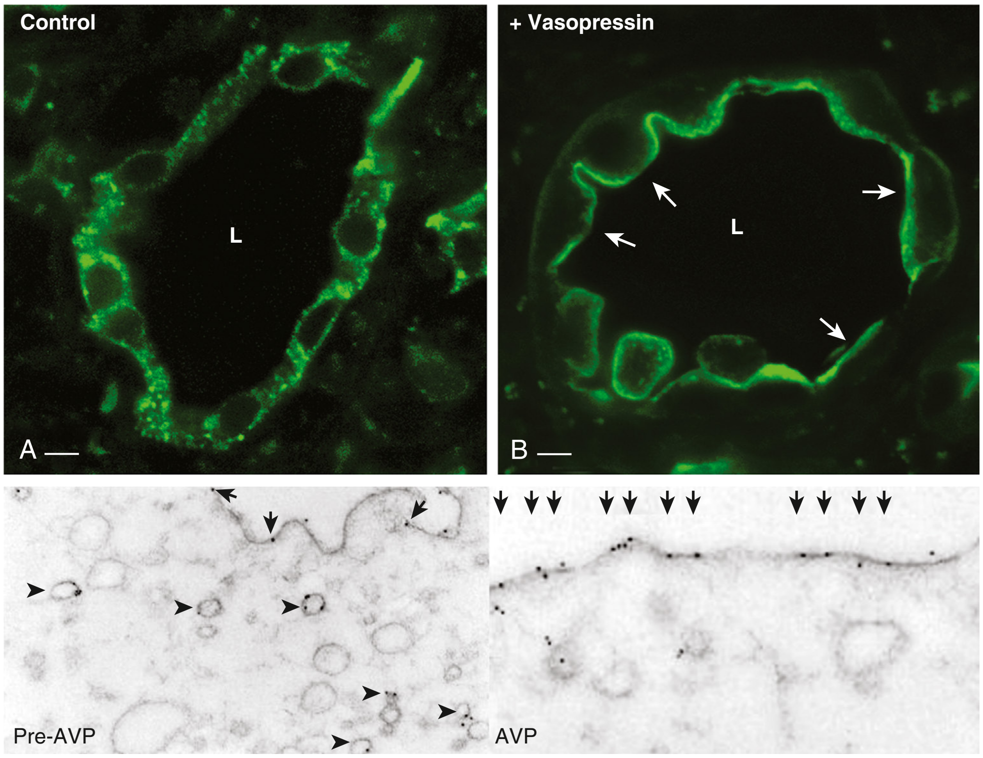

Fig. 10.9 Increased plasma membrane expression of AQP2 in principal cells of vasopressin-deficient Brattleboro rat kidney inner medullary collecting duct injected with vasopressin for 15 minutes. Kidneys were then fixed, sectioned, and immunostained using anti-AQP2 antibodies. Under control conditions (A), AQP2 has a cytosolic distribution in principal cells. After perfusion with vasopressin (AVP) (B), AQP2 shows an increased apical localization in principal cells *(arrows).* A weaker basolateral localization of AQP2 in principal cells is also visible in this section. Lower two panels, the effect of AVP on AQP2 distribution by immunogold electron microscopy. Tubules were perfused with 4 nM DDAVP for 60 minutes Left panel (pre-AVP), the apical region of a principal cell, with gold particles (detecting AQP2) distributed on cytoplasmic vesicles, as well as a few on the apical plasma membrane *(arrows).* After AVP treatment, the number of gold particles on the apical plasma membrane is greatly increased *(arrows),* and the number of labeled cytoplasmic vesicles *(arrowheads)* is decreased. *L,* Tubule lumen. Bar equals 5 μm. (Lower panels adapted from Nielsen S, Chou CL, Marples D, et al. Vasopressin increases water permeability of kidney collecting duct by inducing translocation of aquaporin-CD water channels to plasma membrane. *Proc Natl Acad Sci U S A.* 1995;92:1013–1017.)

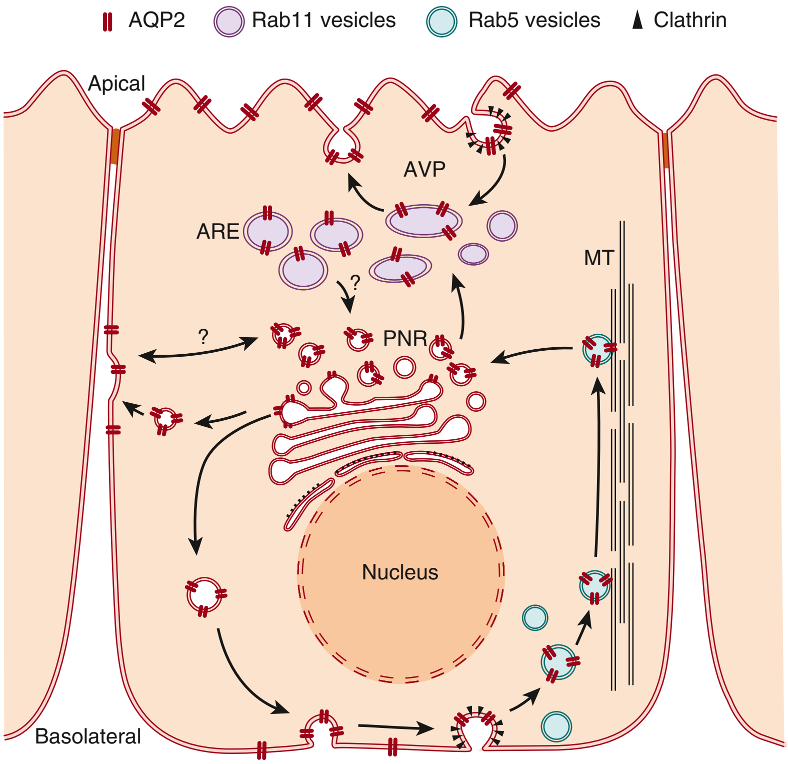

Fig. 10.10 Aquaporin-2 (AQP2) follows a transcytotic pathway before apical membrane delivery. From vesicles in the perinuclear region (PNR), probably originating from the trans-Golgi network, AQP2 can be delivered to the basolateral plasma membrane before reaching the apical surface of epithelial cells. From there, it is retrieved by clathrin-mediated endocytosis into Rab5-positive endosomes *(green)*, which move in a microtubule (MT)-dependent manner to the PNR and ultimately to Rab11-positive apical recycling endosomes (AREs, *purple*). These Rab11-positive vesicles are involved in recycling AQP2 constitutively to and from the apical plasma membrane. The physiologic stimulus, vasopressin (AVP), increases apical AQP2 expression in two ways. It increases exocytosis from the Rab11 compartment and inhibits clathrin-mediated endocytosis of AQP2 from the apical plasma membrane. The delivery of AQP2 to the basolateral membrane of collecting duct principal cells may be important for collecting duct tubulogenesis, whereas apical AQP2 is necessary for urine concentration. (From Yui N, Lu HAJ, Chen Y, et al. Basolateral targeting and microtubule-dependent transcytosis of the aquaporin-2 water channel. *Am J Physiol Cell Physiol.* 2013;304:C38–C48.)

AQP3 is predominant) than in the medullary collecting ducts (where AQP4 is the predominant basolateral water channel).

OVERVIEW OF AQUAPORIN-2 REGULATION BY VASOPRESSIN IN COLLECTING DUCT PRINCIPAL CELLS

The AVP-stimulated increase in collecting duct water permeability and urine concentration correlates with relocalization of AQP2 from intracellular vesicles to the plasma membrane of principal cells (Fig. 10.9).[160,168,203,204] AQP2 accumulation at the cell surface in response to AVP results from increased AQP2 exocytosis and decreased AQP2 endocytosis.[205,206] The relocation is reversible upon AVP removal or in animals either infused with a V_2R antagonist or subjected to water loading to reduce circulating AVP levels.[207-209] Clathrin-coated pits are critical for the internalization of both AQP2 and the V_2R,[117,118,126,210,211] with inhibition of clathrin-mediated endocytosis causing AQP2 plasma membrane accumulation.[162,210,212-217] After AVP removal, part of the internalized AQP2 accumulates in endosomes and is reinserted into the plasma membrane.[167,212,213,218] This likely occurs via classical endosomal recycling compartments[213,219-223] with the vacuolar protein sorting-associated protein 35 (Vps35) playing a role.[224] A significant amount of AQP2 also accumulates in multivesicular bodies (MVBs)[207,225] and is subsequently directed to lysosomes for degradation or recycling compartments, or it is transported to the cell surface via transport vesicles that derive from the MVBs.

The fate of internalized AQP2 seems to be, at least in part, regulated by ubiquitylation of AQP2 at K270,[188,226-228] which enhances AQP2 endocytosis to MVBs and increases lysosomal degradation of AQP2.[228] AVP removal increases ubiquitylated AQP2 levels,[228,229] suggesting that ubiquitylation is a mechanism to reduce collecting duct water reabsorption. Several studies showed that the E3 ubiquitin ligase CHIP is involved in AQP2 ubiquitylation.[230-232] In addition, the E3 ligases NEDD4 and NEDD4L can mediate ubiquitylation and degradation of AQP2 via the NEDD4 family-interacting proteins NDFIP1/2.[233] The ubiquitin-specific protease USP4, which is increased in expression by AVP, is involved in deubiquitylation of AQP2, thereby determining whether AQP2 is recycled to the plasma membrane or targeted for lysosomal degradation.[234] Under certain conditions AQP2 can also be degraded in autophagolysosomes, as demonstrated in several animal models of acquired NDI.[235-238] Some AQP2-containing MVBs fuse with the apical membrane of principal cells and release small nanovesicles known as exosomes into the tubule lumen. These exosomes contain AQP2 on their limiting membranes,[239,240] in addition to a variety of different proteins,[241,242] mRNAs, and microRNAs within their lumen.[243,244] The physiologic relevance of this urinary excretion of AQP2 remains unknown, but the amount of exosomal AQP2 can be increased by AVP[245] and a role in cell–cell communication has been proposed.[244]

AVP also increases AQP2 expression through enhanced AQP2 transcription and reduced degradation,[246,247] the latter likely related to AVP-induced AQP2 membrane retention and reduced lysosomal degradation.[248] A rise in intracellular cAMP and activation of protein kinase A (PKA) are required for AVP-stimulated *AQP2* gene transcription.[249-251] As the *AQP2* promoter contains a cAMP responsive element (CRE), increased *AQP2* transcription has been suggested to be via PKA-mediated phosphorylation of the transcription factor CREB.[252,253] However, some studies have suggested that CREB effects are likely to be indirect.[254] Other transcription factors involved in *AQP2* expression include NFAT family members, YAP, Elf3 and Elf5, GATA2 and GATA3, and NF-κB.[255-263] *AQP2* expression is also influenced by histone H3 lysine 27 (H3K27) acetylation, a modification associated with open chromatin and increased transcription. V_2R activation increases H3K27 acetylation across the *AQP2* promoter,[251] while it is decreased in hypokalemia-induced NDI with low *AQP2* expression.[264] Epigenetic control or transcriptional regulation of *AQP2* abundance is also affected by miRNAs, short RNA molecules that act as posttranscriptional regulators of gene expression by blocking protein translation and/or inducing mRNA degradation.[265-267]

MECHANISMS OF AQUAPORIN-2 TRAFFICKING

A wealth of information exists regarding the regulated trafficking, function, structure, and water transport capacity of AQP2.[151,203,221,250,268-286] Following is a discussion of some mechanisms of AQP2 trafficking that are continually evolving in parallel with new discoveries related to the targeting and trafficking of membrane proteins in general.

ROLE OF THE CYTOSKELETON IN AQUAPORIN-2 TRAFFICKING

Actin associates directly with AQP2[287-289] or AQP2-containing vesicles.[290] Upon AVP-mediated actin depolymerization, AQP2 accumulates in the plasma membrane.[291-294] Apical fluid shear stress also depolymerizes the apical actin cytoskeleton and causes AQP2 membrane accumulation.[295,296] A role for A-Kinase Anchoring Protein 220 (AKAP220), AKAP13, and Rho GTPases in modulating the actin effects on AQP2 has been proposed.[293,297-302] AQP2 also complexes with various other actin-associated proteins including myosins,[290,294,303-305] Rab proteins,[222,306] members of the ezrin-radixin-moesin (ERM) family,[307,308] and the signal-induced proliferation-associated gene 1 (SPA-1).[301] AVP only induces significant actin depolymerization in cells expressing AQP2,[296] suggesting phosphorylation of AQP2 changes its binding to different components of the actin cytoskeleton and critically regulates local actin reorganization to initiate its movement.[288] The integrin-linked kinase (ILK) is also important in orchestrating cytoskeletal organization during AQP2 recycling and entry into the exocytotic pathway,[309,310] while the actin-related protein Arp2/3 is essential for delivery of AQP2 to the plasma membrane.[311]

Dynein and dynactin, a protein complex linking microtubules and vesicles, are associated with AQP2-bearing vesicles.[312] Depolymerization of microtubules partially inhibits AVP-induced osmotic water permeability in target epithelia[313-315] and apical localization of AQP2.[160,285,316,317] A role of microtubules in the basolateral to apical transcytosis of AQP2 has also been suggested (see Fig. 10.10).[178] Together, the data on microtubules indicate that they are predominantly responsible for long-range trafficking of AQP2 vesicles toward the plasma membrane and localization of AQP2 inside the cell after internalization, but that the final steps of vesicle approach and fusion are microtubule independent.[318]

A variety of SNARE proteins colocalize with AQP2 in principal cells. Of these, VAMP-2 (vesicle-associated membrane protein 2, synaptobrevin-2), VAMP-3 (cellubrevin), VAMP-8, syntaxin 3, and SNAP23 (synaptosomal-associated protein 23) are important for AQP2 trafficking.[319-321] Interaction of AQP2 and the SNARE complex may be mediated by the protein snapin [322] and/or by the angiotensin-converting enzyme 2 homolog collectrin.[323]

ESSENTIAL ROLE OF AQUAPORIN-2 PHOSPHORYLATION

The rise in intracellular cAMP levels following V_2R stimulation and subsequent activation of PKA are important for AQP2 trafficking by affecting the phosphorylation status of AQP2.[251,324] That process is supported by PKA-anchoring proteins[325-329] and phosphatase inhibitors altering cell-surface accumulation of AQP2.[330,331] However, V_2R-mediated increases in cAMP are not absolutely necessary for AQP2 membrane targeting and alternative pathways exist.[332-335]

AQP2 contains several phosphorylation sites for protein kinases,[136,328,336,337] some of which are important for AQP2 trafficking.[188,226-228] Whether any of the phosphorylation sites modulate AQP2 unit water permeability is controversial.[324,338,339] Early work focused on the involvement of S256 phosphorylation in AQP2 trafficking, with the current consensus being that S256 phosphorylation is necessary for AVP-induced cell-surface *accumulation* of AQP2.[186,340-344] The importance of this site is highlighted by a mutation that destroys the PKA phosphorylation site at S256, resulting in NDI in humans.[344,345] The roles of S261, S264, and S269 (threonine in humans) are slowly being revealed.[338] All three phosphorylated forms are localized to some degree in the plasma membrane in vivo.[225,346-348] While AVP increases phosphorylation at S256, S264, and S269, AVP decreases the abundance of pS261, whereas AMP-activated kinase (AMPK) activation increases pS261 levels.[328,336,349] S261 phosphorylation follows AQP2 ubiquitylation and appears to play an important role in the sorting of AQP2 to degradation pathways and in some situations basolateral targeting of AQP2.[350,351] Interestingly, activation of AMPK with metformin increases AQP2 phosphorylation in general,[352] whereas levels of phosphorylation are significantly attenuated under acidic conditions.[353] The pS269 form of AQP2 is predominantly detected in the apical plasma membrane, but S269 phosphorylation can occur in recycling endosomes before reaching the membrane.[225,348,354,355] Phosphorylation of S269 overrides the effect of AQP2 ubiquitylation on channel endocytosis.[227,346,347]

ROLE OF PHOSPHORYLATION IN EXOCYTOSIS AND ENDOCYTOSIS OF AQUAPORIN-2

Although S256 phosphorylation is necessary for AVP-induced cell-surface *accumulation* of AQP2, the role of phosphorylation in AQP2 exocytosis is complex. In the absence of AVP, both wild-type AQP2 and an AQP2 S256A

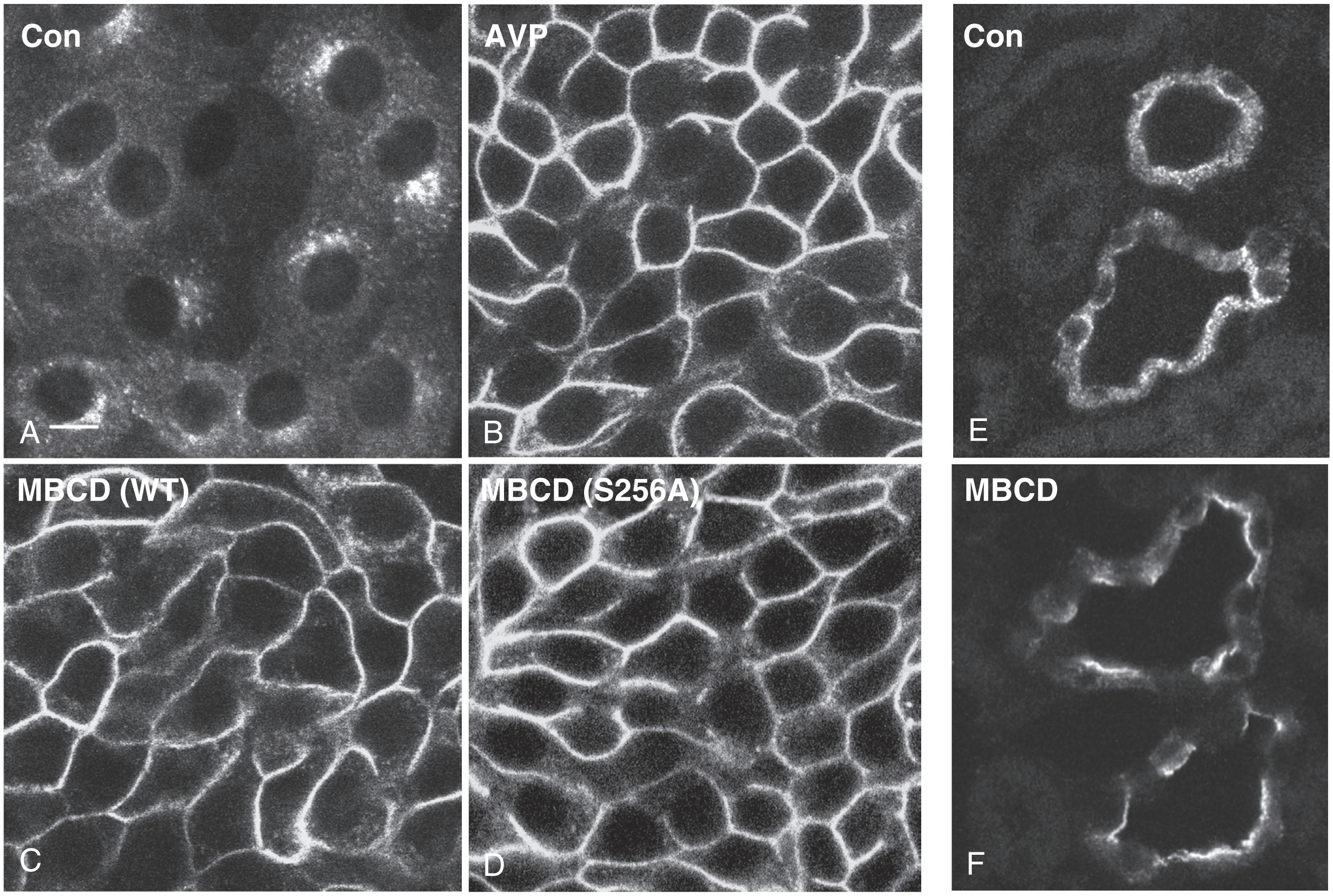

Fig. 10.11 Methyl-β-cyclodextrin (MBCD) stimulates aquaporin-2 (AQP2) membrane accumulation in LLC-PK_1 cells (A to D) and collecting duct principal cells in situ (E and F). Immunofluorescence staining for AQP2 in LLC-PK_1 cells expressing wild-type AQP2 (A to C) or a mutant in which the S256 residue has been replaced by alanine (S256A) (D). Under baseline conditions, wild-type AQP2 is located mainly on intracellular vesicles, often concentrated in the perinuclear region of the cell (A). After vasopressin (AVP) treatment, wild-type AQP2 relocates to the plasma membrane (B). When endocytosis is inhibited by application of the cholesterol-depleting drug MBCD, both wild-type and S256A AQP2 accumulate at the cell surface in the absence of vasopressin (C and D). This result shows that both wild-type AQP2 and S256A AQP2 are constitutively recycling between intracellular vesicles and the plasma membrane and that inhibiting endocytosis with MBCD is sufficient to cause membrane accumulation, even in the absence of S256 phosphorylation of AQP2. In collecting duct principal cells (inner stripe of outer medulla) in situ, AQP2 is located on vesicles scattered throughout the cytoplasm after perfusion of intact kidneys in vitro (E). However, after perfusion of kidneys for 60 minutes with 5 mmol/L MBCD, increased apical plasma membrane expression of AQP2 is seen (F). This finding indicates that AQP2 is constitutively recycling through the apical plasma membrane in principal cells in situ and that membrane accumulation can be induced by blocking endocytosis (with MBCD) even in the absence of vasopressin. *Con,* Control.

mutant protein accumulate on the plasma membrane upon inhibition of endocytosis using the cholesterol-depleting drug methyl-β-cyclodextrin (MBCD) (see Fig. 10.11).[214] This indicates that AQP2 that is not phosphorylated at S256 can constitutively recycle to and from the plasma membrane. AVP also increases exocytosis of vesicles in AQP2-expressing cells whether or not AQP2 is phosphorylated at S256.[205] Thus AVP-induced cell-surface *accumulation* of AQP2 requires S256 phosphorylation and AQP2 is present in "endocytosis-resistant" membrane domains after AVP treatment.[206,214,356] Exocytotic *insertion* of AQP2 into the plasma membrane is probably independent of this phosphorylation event, however. S256 phosphorylation slows AQP2 internalization[248] but does not prevent it.[357,358] A collection of evidence has suggested that phosphorylation-mediated interaction of AQP2 with other regulatory proteins is important for modulating cell-surface accumulation of AQP2. For example, AQP2 phosphorylation at S256 or S269 modifies its interaction with key proteins of the vesicle docking/fusion apparatus or endocytotic machinery.[206,229,248,346,359-361]

VASOPRESSIN-REGULATED UREA TRANSPORT IN THE INNER MEDULLA

ACCUMULATION OF UREA IN RENAL INNER MEDULLA

Urea's importance in the urinary concentrating mechanism has been appreciated since 1934, when Gamble et al initially described "an economy of water in renal function referable to urea." Maximal urine-concentrating ability is lower in protein-deprived or malnourished humans, and urea infusion restores urine-concentrating ability.[363] Urea accumulates within the inner medulla, a process that is partly dependent on variable urea permeabilities along the collecting duct system (Fig. 10.12). The terminal inner medullary collecting duct possesses a high urea permeability,[364] which can be further increased by AVP.[145,365,366] Tubular fluid entering the cortical collecting duct has a relatively low urea concentration. However, during antidiuresis, water is osmotically reabsorbed from the urea-impermeable cortical and outer medullary collecting duct, causing a progressive increase in the luminal urea concentration (Fig. 10.13). Thus when the tubule fluid reaches the highly urea-permeable terminal inner medullary collecting duct (due to the presence of the UT-A1 and UT-A3 urea transporters, see later), urea rapidly exits from the lumen to the inner medullary interstitium, where it is "trapped" by countercurrent urea exchange between descending and ascending flows in both vasa recta and loops of Henle. Under steady-state conditions, and in the continued presence of AVP, urea nearly equilibrates across the inner medullary collecting duct epithelium and thus osmotically balances the urea in the collecting duct lumen, preventing osmotic diuresis (Fig. 10.14).

Within the inner medulla, countercurrent exchange of urea between the descending and ascending vasa recta is facilitated by their proximity and by the extremely high urea permeability of the ascending vasa recta.[80] Indeed, the concentration of urea exiting the inner medulla via the ascending vasa recta is similar to that in the descending vasa recta.[76,365] This minimizes the washout of urea from the inner medulla. However, countercurrent exchange cannot completely eliminate urea loss from the inner medullary interstitium, since the volume flow rate of blood in the ascending vasa recta exceeds that in the descending vasa recta.[367] During antidiuresis, water reabsorbed from both inner medullary collecting ducts and descending limbs is carried away by the ascending vasa recta, resulting in a higher-volume flow rate and an increased mass flow rate of urea. This ensures that the inner medullary vasculature continually removes urea from the inner medulla. While quantitatively the most important loss of urea from the inner medullary interstitium occurs via the vasa recta,[368] that urea loss is limited by urea recycling pathways (Fig. 10.15):

a. Urea recycling through the ascending limbs, distal tubules, and collecting ducts
   Urea that escapes the inner medulla in the ascending limbs of the long loops of Henle is carried back through the thick ascending limbs, distal convoluted tubules, and early portions of the collecting duct system by the flow of tubule fluid.[369] When it reaches the urea-permeable part of the inner medullary collecting ducts, it passively exits into the inner medullary interstitium and repeats the cycle.
b. Urea recycling through the vasa recta, short loops of Henle, and collecting ducts
   Delivery of urea to the superficial distal tubule exceeds delivery out of the superficial proximal tubule.[369-371] This implies that net urea addition occurs somewhere along the short loops of Henle. One possible mechanism is that the urea leaving the inner medulla in the vasa recta is transferred to the descending limbs of the short loops of Henle[370] and carried through the superficial distal tubules back to the urea-permeable part of the inner medullary collecting ducts, where it passively exits, completing the recycling pathway. The physical association between the vasa recta and the descending limbs of the short loops of Henle would facilitate this transfer of urea in the inner stripe of the outer medulla.[54,372] Although the presence of the urea transporter UT-A2 in the thin descending limb of short loops of Henle[7,373] supports this pathway, studies in knockout mice have raised doubts about its importance (discussed later[4,374]).
c. Urea recycling between ascending and descending limbs of the loops of Henle
   The urea permeability of thick ascending limbs from the inner stripe of the outer medulla is low.[375,376] However, the urea permeability of thick ascending limbs from the outer stripe of the outer medulla and the medullary

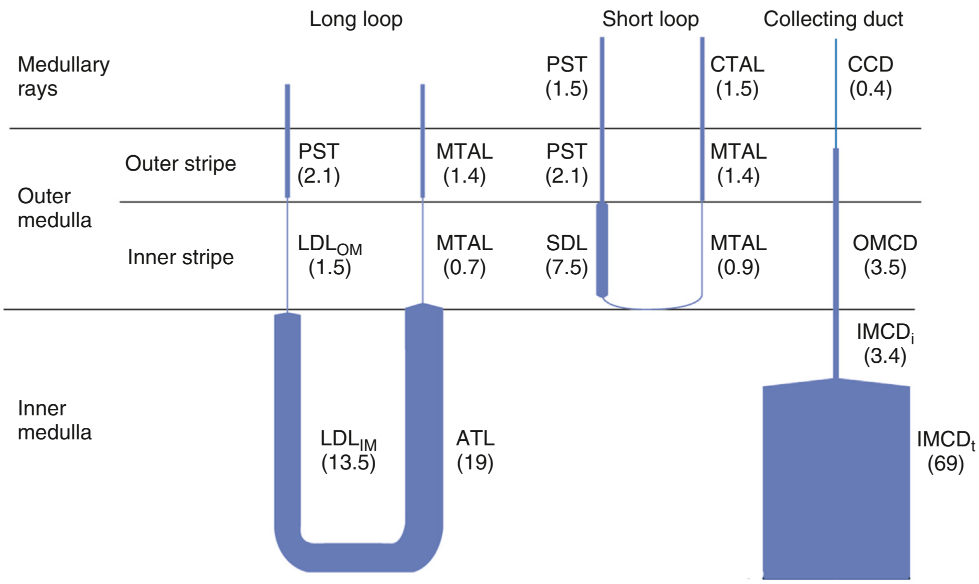

Fig. 10.12 Urea permeabilities of mammalian renal tubule segments. The width of each segment in the diagram is distorted to be proportional to the urea permeability of that segment. Numbers in parentheses are measured values for the permeability coefficient (×10^{-5} cm/sec). Values are from isolated perfused tubule studies. *ATL,* Ascending thin limb; *CTAL,* cortical thick ascending limb; *IMCD$_i$,* initial inner medullary collecting duct; *IMCD$_t$,* terminal inner medullary collecting duct; *LDL$_{OM}$,* thin descending limb of long-looped nephron in outer medulla; *MTAL,* medullary thick ascending limb; *OMCD,* outer medullary collecting duct; *PST,* proximal straight tubule; *SDL,* thin descending limb of short-looped nephron.

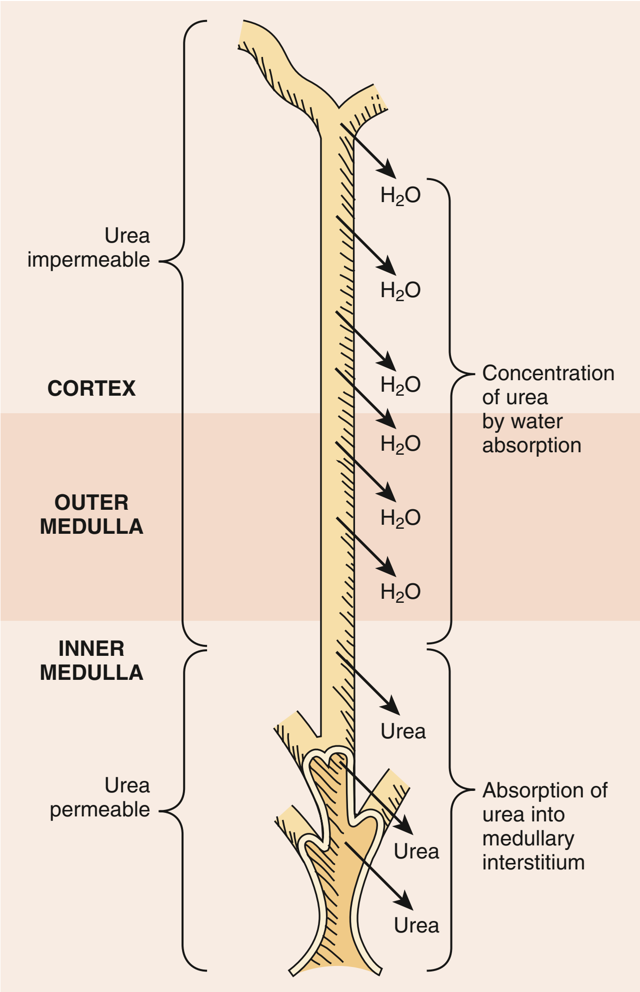

Fig. 10.13 Schematic representation of the mammalian collecting duct system showing principal sites of water absorption and urea absorption. Water is absorbed in the early part of the collecting duct system, driven by an osmotic gradient. Since urea permeabilities of cortical, outer medullary, and initial inner medullary collecting duct are low, the water absorption concentrates urea in the lumen of these segments. When the tubule fluid reaches the terminal inner medullary collecting duct, which is highly permeable to urea, urea rapidly exits from the lumen. This urea is trapped in the inner medulla because of countercurrent exchange.

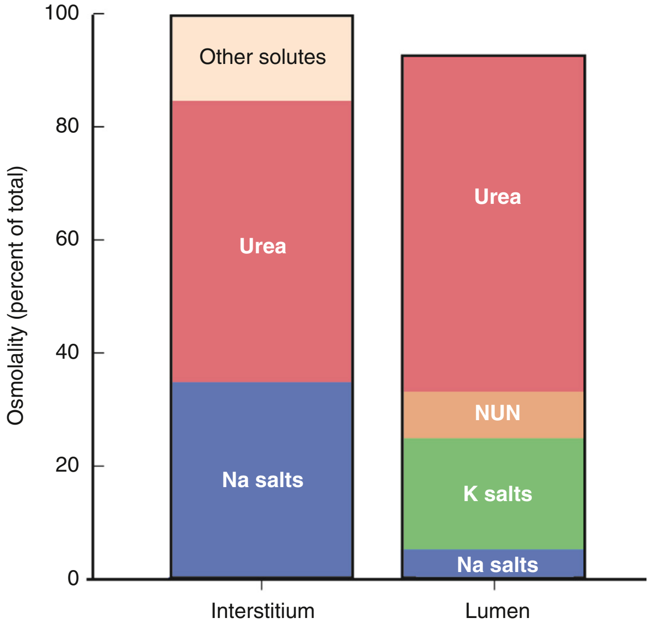

Fig. 10.14 Solutes that account for osmolality of medullary interstitium and tubule fluid in the inner medullary collecting duct during antidiuresis in rats. Urea nearly equilibrates across the inner medullary collecting duct epithelium because of rapid facilitated urea transport. Although the osmolalities of the fluid in the two spaces are nearly equal, the nonurea solutes can differ considerably between the two compartments. Typical values in untreated rats are presented. Values can differ considerably in other species and in the same species with different diets. *NUN,* Nonurea nitrogen.

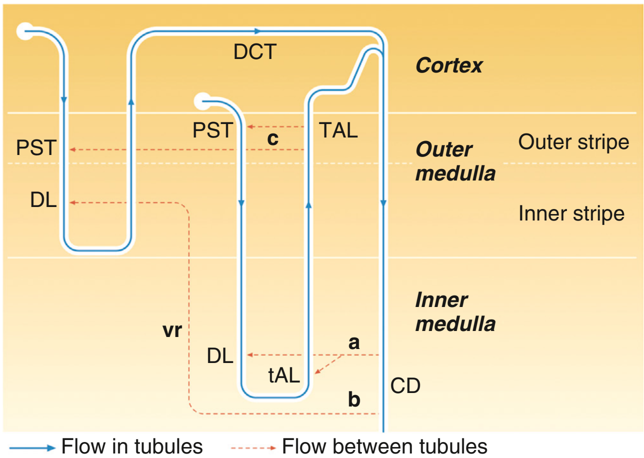

Fig. 10.15 Pathways of urea recycling in renal medulla. *Solid blue lines* represent a short-looped nephron (left) and a long-looped nephron (right). Transfer of urea between nephron segments is indicated by *dashed red arrows* labeled a, b, and c corresponding to recycling pathways described in the text. *CD,* Collecting duct; *DCT,* distal convoluted tubule; *DL,* descending limb; *PST,* proximal straight tubule; *tAL,* thin ascending limb; *TAL,* thick ascending limb; *vr,* vasa recta. (Knepper MA, Roch-Ramel F. Pathways of urea transport in the mammalian kidney. *Kidney Int.* 1987;31:629–633.)

rays is relatively high.[375,377] A urea recycling pathway has been proposed in which urea is reabsorbed from thick ascending limbs and is secreted into neighboring proximal straight tubules, forming a recycling pathway between the ascending limb and descending limbs of the loop of Henle.[2,368] This transfer of urea is likely to depend on a relatively attenuated effective blood flow in these regions. Urea secretion into the proximal straight tubules can occur by passive diffusion,[377] active transport,[378] or a combination of both. The urea that enters the short-looped nephrons will be carried back to the inner medulla by the flow of tubule fluid through the superficial distal tubules and cortical collecting ducts, reentering the inner medullary interstitium by reabsorption from the terminal inner medullary collecting duct. The urea that enters long-looped nephrons returns to the inner medulla directly through the descending limbs of the loops of Henle.[368]

UREA TRANSPORTER PROTEINS

Two urea transporter genes have been cloned in mammals: The UT-A (*Slc14A2*) gene encodes 6 protein and 9 cDNA isoforms,[363] and the UT-B (*Slc14A1*) gene encodes 2 protein isoforms.[379] Urine-concentrating defects exist in UT-A1/A3 knockout,[95] UT-A2 knockout,[374] UT-B knockout,[380-382] UT-A2/UT-B knockout,[383] and all-UT knockout mice.[384]

The UT-A gene has two promoter elements: one upstream of exon 1 and a second that is located within intron 12 and drives the transcription of UT-A2 and UT-A2b[385-388] (reviewed by Sands and Layton[363]). UT-A promoter I contains a tonicity enhancer (TonE) element and hyperosmolality increases its activity.[386,389] UT-A1 is expressed in the terminal inner medullary collecting duct and is detected in the apical plasma membrane[373,387,390] (see Fig. 10.16). UT-A3 is also expressed in the terminal inner medullary collecting duct, primarily in the basolateral plasma membrane [391-393]. UT-A1 interacts with UT-A3.[394] UT-A2 is expressed in thin descending limbs[7,373,390,395] and displays diurnal variation.[396] The UT-B gene has a single promoter that drives protein expression in descending vasa recta and red blood cells (reviewed by Sands and Layton[363]). UT-B is also the Kidd blood group antigen in humans.[397]

AVP increases the phosphorylation and the apical plasma membrane accumulation of UT-A1 and UT-A3 in inner medullary collecting ducts of the rat.[393,398] UT-A1 is phosphorylated at serines 486 and 499.[336,399-401] AVP effects occur through two cAMP-dependent pathways: PKA and Epac (exchange protein activated by cAMP).[401,402] Epac1 maintains inner medullary osmolality through UT-A1 and UT-A3.[403] Aldosterone inhibits AVP-stimulated urea permeability.[404] UT-A1 is dephosphorylated by multiple phosphatases including protein phosphatase 2A and protein phosphatase 2B (calcineurin).[405,406]

Hyperosmolality increases urea permeability in terminal inner medullary collecting ducts of the rat, even in the absence of AVP,[407-409] suggesting that hyperosmolality is an independent activator of urea transport. Hyperosmolality stimulates urea permeability via activation of PKCα and intracellular calcium,[410-413] while AVP stimulates urea permeability via increases in cAMP.[414] Hyperosmolality increases the phosphorylation and plasma membrane accumulation of both UT-A1 and UT-A3.[393,398,415,416] UT-A1 is phosphorylated by PKCα at serine 494.[412,413,417,418] Mice with genetic knockout of PKCα, nuclear factor of activated T cells 5 (NFAT5), hepatocyte nuclear factor-1β, or transcription

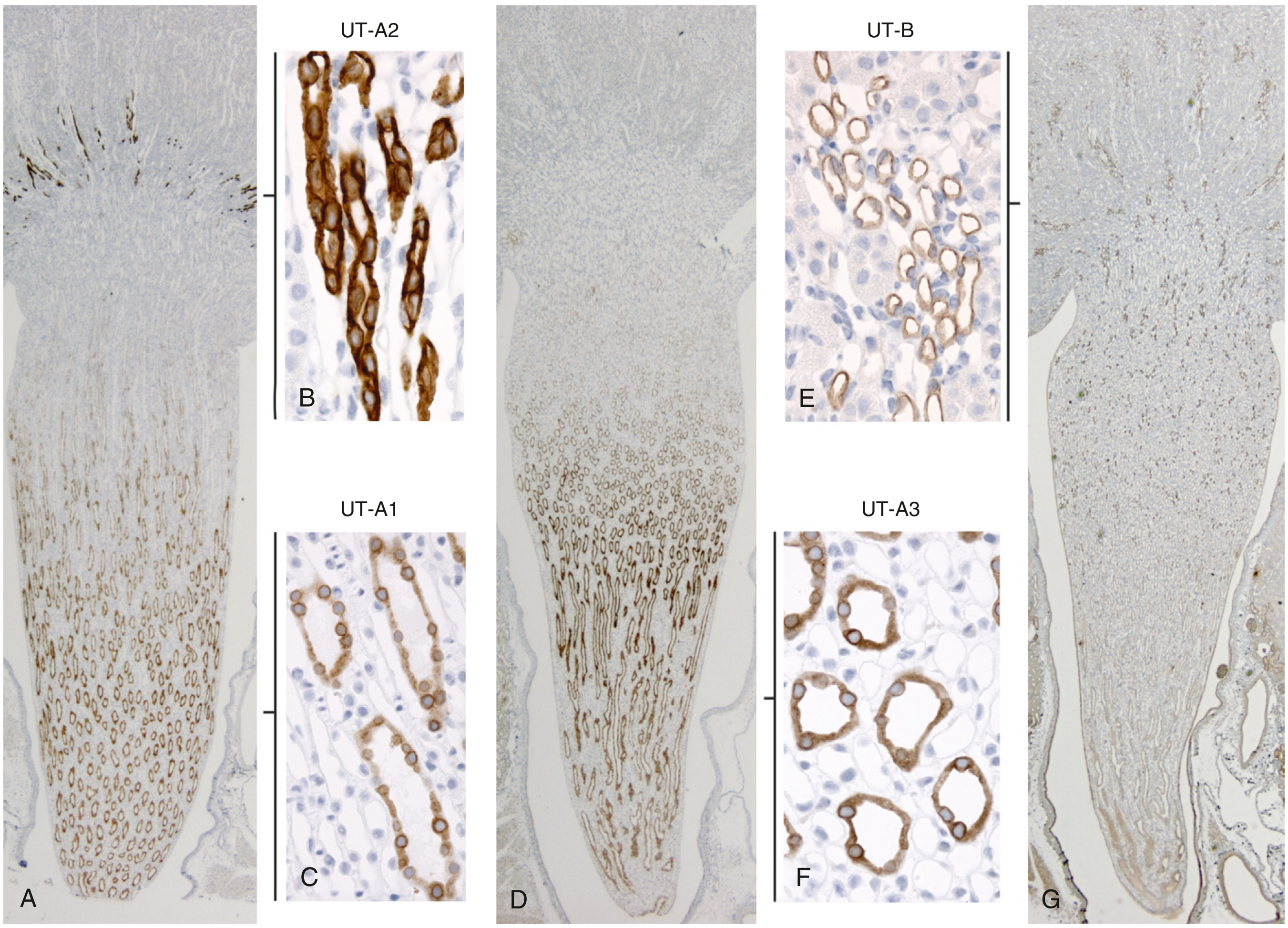

Fig. 10.16 Localization of urea transporters. UT-A1 is localized to the terminal portion of the inner medullary collecting duct, whereas UT-A2 is localized to the thin descending limbs of Henle loop in the inner stripe of outer medulla (A). Higher magnification shows that both UT-A2 (B) and UT-A1 (C) are predominantly intracellular. UT-A3 is localized to the terminal portion of the inner medullary collecting duct (D) and is both intracellular and in the basolateral membrane domains (F). UT-B is expressed in the descending vasa recta (G), where it is localized to the basolateral and apical regions (E). (Adapted from Fenton RA, Knepper MA. Urea and renal function in the 21st century: insights from knockout mice. *J Am Soc Nephrol.* 2007;18: 679–688.)

factors Pax2 and Pax5 have a urine-concentrating defect and reduced levels of UT-A1.[255,413,417,419-424]

Metformin, an AMPK activator, increases UT-A1 and AQP2 phosphorylation, elevates urea and water transport in inner medullary collecting ducts, and enhances urine-concentrating ability in rodent models of congenital NDI.[352,425] A novel AMPK activator, NDI-5033, also increases urine-concentrating ability in rodents.[426] Thus drugs that activate AMPK may be a future therapy for NDI.[352,425-428]

UREA TRANSPORTER KNOCKOUT MICE

Mice with genetic knockout of the two inner medullary collecting duct urea transporters, UT-A1 and UT-A3 (*UT-A1/A3*$^{-/-}$ mice), have complete absence of phloretin-sensitive and AVP-regulated urea transport in their inner medullary collecting duct[95,429-431] Mice lacking only UT-A1 are also unable to increase urine concentration and exhibit impaired urea-selective urine concentration.[432] *UT-A1/A3*$^{-/-}$ mice fed a normal or high-protein diet have a significantly greater fluid intake and urine flow, and after an 18-hour water restriction are unable to reduce their urine flow, resulting in volume depletion and loss of body weight.[95,96] In contrast, on a low-protein diet (4%), *UT-A1/A3*$^{-/-}$ mice do not show polyuria and can reduce their urine flow to a similar level as control mice after water restriction. On a low-protein diet, hepatic urea production is low and urea delivery to the inner medullary collecting duct is low, thus rendering collecting duct urea transport largely immaterial to water balance. Thus the concentrating defect in *UT-A1/A3*$^{-/-}$ mice is due to a urea-dependent osmotic diuresis, compatible with a model of urea handling proposed in the 1950s by Berliner et al[80] Purinergic signaling is enhanced in *UT-A1/A3*$^{-/-}$ mice, which may contribute to their AVP-resistant polyuria.[433] *UT-A1/A3*$^{-/-}$ mice also show reduced fibrosis following unilateral ureteral obstruction.[434]

UT-A1/A3$^{-/-}$ mice have been used to study the "passive mechanism for urine concentration models" for concentration of Na^+ and Cl^- in the inner medulla in the absence of active transport[435,436] (see later). In these models, the passive electrochemical gradient that drives Na^+ and Cl^- exit from the thin ascending limb is indirectly dependent on rapid reabsorption of urea from the inner medullary collecting duct. However, despite a profound decrease in inner medullary urea accumulation in *UT-A1/A3*$^{-/-}$ mice, independent studies failed to demonstrate the predicted decline in Na^+ and Cl^- concentrations in the inner medulla.[95,362,430,431] On the basis of these results, the passive concentrating model in the form originally proposed in 1972 does not appear to be the mechanism by which NaCl is concentrated in the inner medulla. However, mathematical modeling analysis of these same data concluded that the results found in the *UT-A1/A3*$^{-/-}$ mice are consistent with what one would predict for the passive mechanism.[437] Thus the issue remains unresolved at present.

Another hypothesis regarding urea and the urinary concentrating mechanism has been referred to as the "Gamble phenomenon."[362] Gamble and colleagues[362] described that 1. the water requirement for excretion of urea is less than for excretion of an osmotically equivalent amount of NaCl; and 2. less water is required for the excretion of urea and NaCl together than the water needed to excrete an osmotically equivalent amount of either urea or NaCl alone. In *UT-A1/A3*$^{-/-}$ mice, both elements of the Gamble phenomenon were absent, indicating that inner medullary collecting duct urea transporters play an essential role.[438] When wild-type mice were given progressively increasing amounts of urea or NaCl in the diet, both substances induced osmotic diuresis but at different excretion levels. Mice were unable to increase urinary NaCl concentrations above 420 mM. Thus the second component of the Gamble phenomenon derives from the fact that both urea and NaCl excretion are saturable, presumably resulting from an ability to exceed the respective reabsorptive capacity for urea and NaCl, rather than a specific interaction of urea transport and NaCl transport at an epithelial level.

Mice lacking UT-A3, but expressing UT-A1, have normal basal urea permeability in the inner medullary collecting duct, but unlike control mice, AVP did not stimulate urea permeability above basal levels.[439] Surprisingly, urine-concentrating ability in the mice lacking only UT-A1 was similar to controls.[439] Mice lacking UT-A1 and UT-A3, but not UT-A3 alone, have elevated medullary nitric oxide production and salt wasting.[440]

Mice lacking UT-A2 have reduced urine-concentrating ability.[431,441] The urine-concentrating defect is thought to result from impairment of urea recycling.[431,441] UT-B knockout mice also have a reduced urine-concentrating ability that is similar to humans lacking UT-B.[431,441] As noted earlier, UT-B is the Kidd blood group antigen and people lacking the Kidd antigen are unable to concentrate their urine above 800 mOsm/kg H_2O, even after overnight water deprivation or exogenous AVP administration.[442] Unexpectedly, UT-A2 deletion in UT-B knockout mice partially corrected the concentrating defect observed in mice lacking only UT-B.[383] These results suggest that rather than playing a role in maintaining urea concentration during the normal steady state, UT-A2 may function to move urea during the acute transition from diuresis to antidiuresis.[383]

Mice lacking all urea transporters have a high output of dilute urine and reduced blood pressure, and they do not increase urine osmolality following water restriction, acute urea loading, or a high-protein intake.[384] These knockout mice do not exhibit physiologic abnormalities in extrarenal tissues [384] (see "Clinical Relevance: Urearetics").

Clinical Relevance: Urearetics

Urea transporter inhibitors have been developed as potential novel diuretics.[488-490] Dimethylthiourea (DMTU), a urea analog, inhibits UT-A1 and UT-B, results in a sustained and reversible reduction in urine osmolality, an increase in urine volume, and mild hypokalemia in rats.[491,492] Another class of inhibitors, an indole thiazole or γ-sultambenzosulfonamide, is selective for UT-A and results in diuresis with more urea than salt excretion in rats.[493-496] The thienoquinolin PU-48 results in a diuresis in both control and UT-B knockout mice, indicating that it inhibits UT-A. It also reduces urea permeability in perfused rat inner medullary collecting ducts.[497] Since the diuresis induced by PU-48 did not change serum sodium, chloride, or potassium levels, it supports the hypothesis that an agent that targets UT-A1, which is expressed in the last portion of the inner medullary collecting duct, may have less risk for side effects, such as hypokalemia, than conventional diuretics that act in more proximal portions of the nephron.[497] Several urearetics are under development, but none are currently available for human use.[498]

URINE CONCENTRATION AND DILUTION PROCESSES ALONG THE MAMMALIAN NEPHRON

SITES OF URINE CONCENTRATION AND DILUTION

Micropuncture studies of the mammalian nephron have determined the major sites of tubule fluid concentration and dilution (see Fig. 10.7). Regardless of whether the kidney is diluting or concentrating the urine, proximal tubule fluid is always isosmotic with plasma.[443] During water diuresis, the fluid in the distal tubule fluid is hypotonic. During antidiuresis, the fluid in the distal tubule becomes isosmotic with plasma and the osmolality gradually rises between the end of the late distal tubule and inner medullary collecting ducts. Thus the loop of Henle is the major site of dilution of tubule fluid, and dilution processes in the loop occur regardless of whether the final urine is dilute or concentrated. During water diuresis, further dilution of the tubule fluid can occur in the collecting ducts,[444] whereas during antidiuresis, the collecting duct system is the chief site of urine concentration.

MECHANISM OF TUBULE FLUID DILUTION

Micropuncture studies demonstrated low luminal NaCl concentration in the early distal tubule[445] and established the mechanism of tubule fluid dilution in thick ascending limbs.[446,447] NaCl is rapidly reabsorbed by active transport, which lowers the luminal osmolality and NaCl concentration to levels below those in the peritubular fluid. The osmotic water permeability of the thick ascending limb is low, which prevents dissipation of the transepithelial osmolality gradient by water flux. The tubule fluid remains hypotonic throughout the distal tubule and collecting duct system during water diuresis when circulating levels of AVP are low. Even though the tubule fluid remains hypotonic in the collecting duct system, the solute composition of the tubule fluid is modified within the collecting duct, mainly by Na^+ absorption and K^+ secretion. Active NaCl reabsorption from the cortical collecting duct results in a further dilution of the collecting duct fluid, beyond that achieved in the thick ascending limbs.[444]

MECHANISM OF TUBULE FLUID CONCENTRATION

When circulating AVP levels are high, net water absorption occurs between the late distal tubule and collecting ducts.[369] Since water is absorbed in excess of solutes,[448] collecting duct fluid is concentrated chiefly by water absorption rather than by solute addition.

The driving force for water absorption is provided by an axial osmolality gradient in the renal medullary tissue, with an increasing gradient along the outer and inner medulla and the highest degree of hypertonicity at the papillary tip.[449] In addition, within the medulla the osmolality of the collecting duct tubule fluid is as high as in the loops of Henle and the osmolality of vasa recta blood near the papillary tip is virtually equal to that of the final urine.[449] Micropuncture studies by Gottschalk and Mylle[443] confirmed that the osmolality of the fluid in the loops of Henle, the vasa recta, and the collecting ducts is approximately the same (see Fig. 10.7), supporting that collecting duct fluid is concentrated by osmotic equilibration with a hypertonic medullary interstitium. Indeed, collecting ducts have a high water permeability in the presence of AVP,[365,450] as is required for osmotic equilibration.

How is the corticomedullary osmolality gradient generated? The principal solutes responsible for the osmolality gradient are NaCl and urea[451] (Fig. 10.17). The increase in the NaCl concentration gradient along the corticomedullary axis occurs predominantly in the outer medulla, with only a small increase in the inner medulla. In contrast, the increase in urea concentration occurs predominantly in the inner medulla, with little or no increase in the initial outer medulla.

GENERATION OF THE AXIAL SODIUM CHLORIDE GRADIENT IN THE RENAL OUTER MEDULLA

In both diuresis and antidiuresis, an osmolality gradient is maintained along the cortico-medullary axis of the outer medulla (see Fig. 10.17)[452] that arises mostly from an accumulation of NaCl. Because the outer medullary collecting duct is water permeable to varying degrees in diuresis and antidiuresis, the accumulation of NaCl in the outer medulla cannot depend on a sustained osmolality difference across the collecting duct epithelium. Instead, the concentrating mechanism must depend on the loops of Henle, the vasculature, and their interactions.

In 1942, Kuhn and Ryffle[453] proposed a paradigm based on countercurrent multiplication. Osmotic pressure is raised along parallel but opposing flows in nearby tubes that are made contiguous by a hairpin turn (Fig. 10.18): A transfer of solute from one tubule to another (i.e., a single effect) would augment (multiply) the osmotic pressure in the parallel flows. Thus by means of the countercurrent configuration, a small transverse osmotic difference would be multiplied into a relatively large difference along the axes of flow. In 1951, Hargitay and Kuhn[454] identified that the loop of Henle had parallel tubes joined by a hairpin turn. Thus the loops of Henle were proposed as the source of the outer medullary gradient, and that gradient was hypothesized to draw water out of water-permeable collecting ducts. Kuhn and Ramel[455] subsequently used a mathematical model to show that active transport of NaCl from thick ascending limbs could serve as the single effect, which was confirmed experimentally alongside osmotic absorption of water from collecting ducts.[369,446-448] Further experiments indicating high water permeability in descending limbs of short[456] and long loops[8,9,11] suggested that the accumulation of NaCl from thick limbs concentrated descending limb tubular fluid by osmotic water withdrawal, rather than by NaCl addition (Fig. 10.18).

The paradigm of countercurrent multiplication has evolved alongside advanced anatomic details of the medulla. In particular, the descending limbs of short loops are anatomically separated from ascending limbs, with inner stripe portions of short loops near (or within) the vascular bundles and thick limbs near the collecting ducts.[51,457] This configuration is inconsistent with direct interactions between counter-flowing limbs. Furthermore, the absence of AQP1 from descending limbs of short loops in the inner

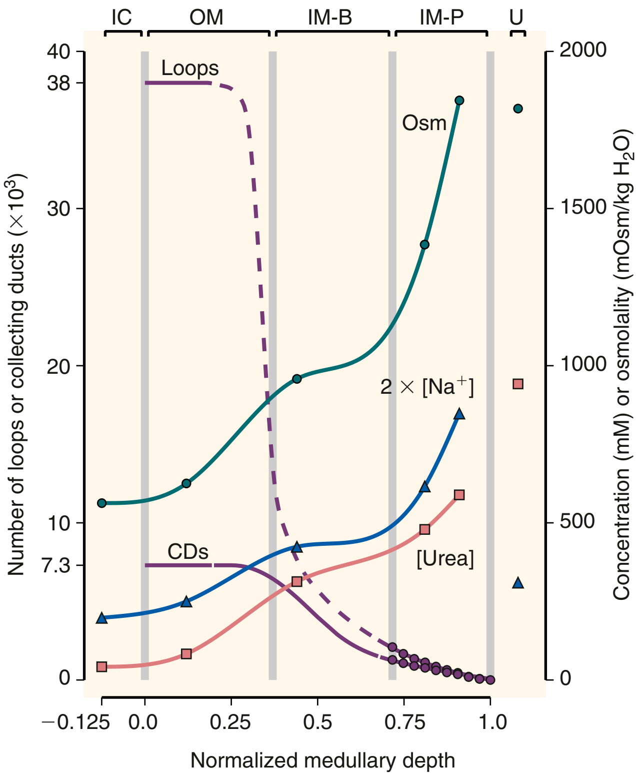

Fig. 10.17 Axial gradients of solutes in relation to nephron segments in the rat kidney in an antidiuretic state. Osmolality, urea concentration, and sodium concentration (plus accompanying anion) are shown (scale at right); also, loop of Henle and collecting duct populations (scale at left). *IC,* Inner cortex; *IM,* outer part (base) of inner medulla; *OM,* outer medulla; *P,* papilla or inner part (tip) of inner medulla; *U,* urine. The density of loops of Henle and collecting ducts decreases in inner medulla because collecting ducts merge and loops turn back. The axial osmolality gradient is steeper in the outer medulla and papilla than in the outer part of the inner medulla. The gradient is steepest in the papilla, where the osmolality and concentration profiles appear to increase exponentially. The shape of the sodium profile has been corroborated by electron microprobe measurements.[499] Figure is based on published data. Curves connecting data points are natural cubic splines.[500] Dashed curve segments are interpolations without supporting measurements. Tubule populations in papilla are from reference 501; tubule populations in outer medulla are based on estimates in reference 49. Concentrations and osmolalities are from tissue slices and urine samples collected 4.5 hours after onset of vasopressin infusion at 15 μU/min per 100 g body weight and are derived from figure 5 in reference 502 and figures 1, 3, 9 in reference 452; slice locations were given in reference 8. The osmolality reported in the inner cortex seems high relative to the reported plasma concentration of 314 mOsm/kg H_2O. The osmolality and concentration profiles, as drawn in reference 452, apparently do not take into account relative distances between tissue sample sites. (From Sands JM, Layton HE. The urine-concentrating mechanism and urea transporters. In: Alpern RJ, Caplan MJ, Moe OW, eds. *The Kidney: Physiology and Pathophysiology.* 5th ed. San Diego: Academic Press; 2013:1463–1510)

stripes of mice, rats, and humans[7,17] suggests that the assumption of high water permeability in descending limbs of short loops merits further experimental study.

From these considerations, it seems reasonable to hypothesize that the outer medullary osmolality gradient arises principally from vigorous active transport of NaCl, without accompanying water, from the thick ascending limbs of short- and long-looped nephrons. In rats and mice, the thick limbs are localized near the collecting ducts[458] and mathematical models suggest that at a given level of the outer medulla, the interstitial osmolality will be higher near the collecting ducts than near the vascular bundles.[459,460] This higher osmolality will facilitate water withdrawal from the descending limbs of long loops and from collecting ducts. The ascending vasa recta will act as the collectors of any NaCl that is absorbed from loops of Henle and water that is absorbed from the descending limbs of long loops and from collecting ducts. The countercurrent configuration of the ascending vasa recta, relative to the descending limbs and collecting ducts, is likely to participate in sustaining the axial gradient: As ascending vasa recta fluid ascends toward the cortex, its osmolality will exceed that in the descending limbs of long loops and in the collecting ducts. Thus ascending vasa recta fluid will be progressively diluted as that fluid contributes to the concentrating of fluid in descending limbs of long loops and in collecting ducts, by giving up NaCl to, and absorbing water from, the interstitium (Fig. 10.19).

While the summary above accounts for the elevation of osmolality in the outer medulla, the increasing osmolality gradient along the outer medulla as a function of increasing medullary depth requires countercurrent multiplication, with greater and faster NaCl absorption from thick limbs at deeper levels due to a higher Na-K-ATPase activity.[461] Moreover, because of the water already absorbed in the upper outer medulla, the load of water presented to the thick limbs deep in the outer medulla by descending limbs of long loops and by the collecting ducts is greatly reduced. One caveat of these mechanisms is that our understanding of the outer medulla is mostly based on information obtained from laboratory animals, especially rats and mice, but outer medullary function and structure are likely to vary substantially in other species.

AN UNRESOLVED QUESTION: CONCENTRATION OF SODIUM CHLORIDE IN THE RENAL INNER MEDULLA

The ascending limbs of loops of Henle that reach into the inner medulla are thin walled and do not actively transport NaCl[365,462,463]; nonetheless, in antidiuresis, a substantial axial osmolality gradient is generated in the inner medulla (see Fig. 10.17). For decades, controversy has persisted regarding the nature of the mechanism that generates the inner medullary osmolality gradient and the energy source for the concentrating of nonurea solutes in the inner medullary interstitium. Three major hypotheses have been proposed:

1. **The "Passive Mechanism."** Kokko and Rector [435] and Stephenson [436] independently proposed the "passive mechanism," which depends on the separation of urea and NaCl due to NaCl absorption from the thick ascending limbs. This absorption is the hypothesized energy source for the passive mechanism. In this model, rapid urea reabsorption from the inner medullary collecting duct generates and maintains a high urea concentration in the inner medullary interstitium, causing the osmotic withdrawal of water from the thin descending limb. This concentrates NaCl in the *descending* limb lumen and results in

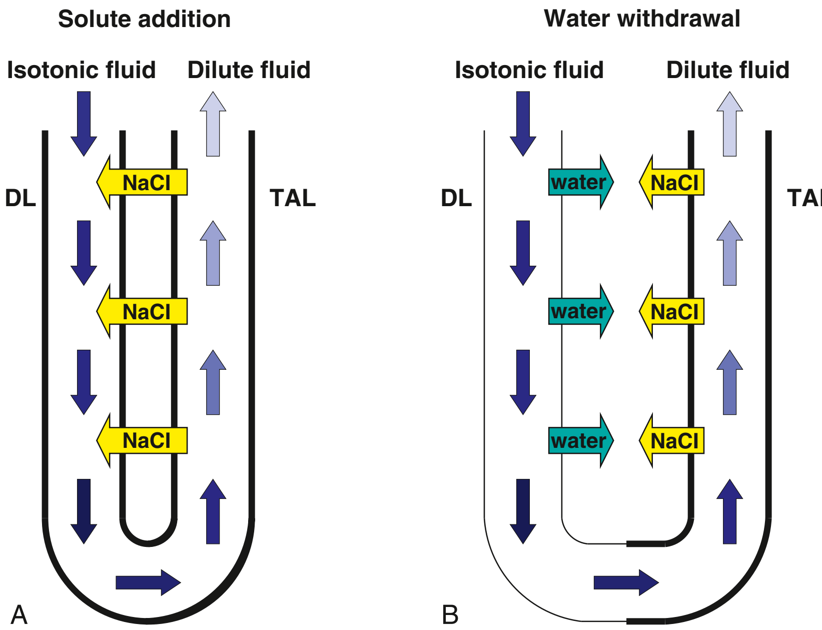

Fig. 10.18 Theoretical mechanisms of countercurrent multiplication in the loop of Henle. (A) Countercurrent multiplication by means of NaCl transfer from an ascending flow to a descending flow. (B) Countercurrent multiplication by means of water withdrawal from a descending flow. NaCl transport from the ascending flow into the interstitium raises interstitial osmolality; this results in passive water transport from the descending flow, which has lower osmolality than the interstitium. In both panels, tubular fluid flow direction is indicated by blue arrows; increasing osmolality is indicated by darkening shades of blue. *DL,* Descending limb of Henle loop; *TAL,* thick ascending limb of Henle loop. Thick black lines indicate that a tubule is impermeable to water; thin lines indicate high permeability to water.

a transepithelial gradient favoring the passive reabsorption of NaCl from the thin *ascending* limb of the Henle loop. Additionally, if the ascending limbs have extremely low urea permeability, then any NaCl that has been reabsorbed from the thin ascending limb will not be replaced by urea. Thus the ascending limb fluid will be dilute relative to the fluid in other nephron segments generating a "single effect" analogous to active NaCl absorption from thick ascending limbs. This single effect can then be multiplied by the counterflow between the ascending and descending limbs of the Henle loops. This model requires that the thin descending limbs are highly permeable to water but not NaCl or urea, whereas the thin ascending limb would have to be permeable to NaCl but not water or urea. However, contrary to the permeability requirements of the passive model, high urea permeabilities have been measured in the thin descending limb and thin ascending limb,[464] whereas little or no osmotic water permeability has been measured in the lower portions of thin descending limbs in the inner medulla.[20] In addition, studies in UT-A1/A3 urea transporter knockout mice found that urea accumulation in the inner medulla was largely eliminated, but inner medullary NaCl accumulation was not affected.[94-96]

Layton and colleagues[19,465-468] reevaluated the passive mechanism by incorporating measured loop NaCl, urea, and water permeabilities, as well as the three-dimensional architecture of the renal medulla, into a detailed mathematical model. These studies suggest that water absorption from descending limbs is not necessary to generate an osmolality gradient and that the urea-permeable loops of Henle can serve as an effective countercurrent urea exchanger in generating a moderately concentrated urine.

2. **Concentrating Mechanism Driven by External Solute.** Jen and Stevenson[469] proposed that the concentrating mechanism of the inner medulla depends on a solute other than NaCl and urea. They demonstrated, in principle, that the continuous addition of small amounts of an unspecified, but osmotically active, solute to the inner medullary interstitium could produce a substantial axial osmolality gradient. Such a solute would have to be generated in the inner medulla by a chemical reaction that produces more osmotically active particles than it consumes. The mechanism of concentration is similar to that driven by urea in the "passive" models[435,436]: The thin descending limbs in the inner medulla are assumed impermeable to the solute (thus it is an "external" solute), and as a result, water is withdrawn from the descending limbs and the concentration of NaCl is raised in descending limb tubular fluid. Beginning at the loop bend, elevated NaCl concentration within the loop will result in a substantial NaCl efflux that will dilute the ascending flow and that is sufficient to generate the axial gradient. In further modeling studies, lactate, generated by anaerobic glycolysis (the predominant means of ATP generation in the inner medulla), was proposed as the solute with two lactate ions generated per glucose consumed.[470,471] However, models developed by Zhang and Edwards[472] and Chen and colleagues[473] predicted that vascular countercurrent exchange would tend to restrict significant glucose availability into the outer medulla and the upper inner medulla, thus limiting the rate of lactate generation in the deep inner medulla where the highest osmolalities are found.
3. **Hyaluronan as a Mechano-osmotic Transducer.** Hyaluronan (or "hyaluronic acid") is a glycosaminoglycan[474,475] that is abundant in the interstitium of the renal inner medulla.[476,477] The hyaluronan in the inner medulla is produced by type 1 interstitial cells, which form characteristic "bridges" between the thin limbs of Henle and the vasa recta.[478] These bridges may delimit, above and below, the nodal compartments identified by Pannabecker and Dantzler.[56] Thus the inner medullary interstitium may be considered to be composed of a compressible, viscoelastic, hyaluronan matrix.

Schmidt-Nielsen suggested that compression of the hyaluronan matrix stores some of the mechanical energy

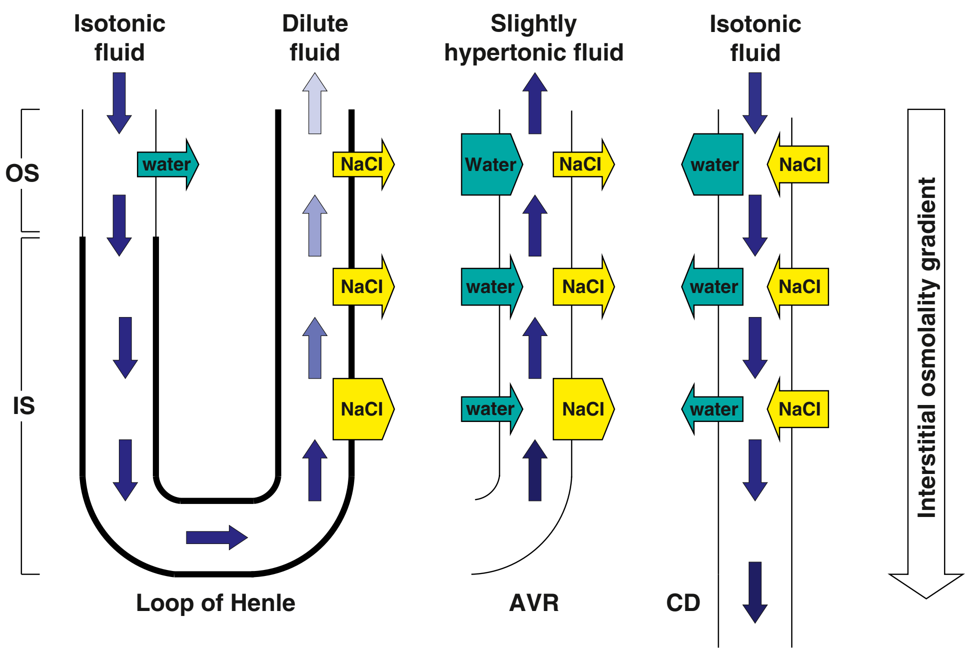

Fig. 10.19 Outer medullary concentrating mechanism based on NaCl addition to the interstitium but without water absorption from descending limbs of short loops. *Arrows* indicate water *(teal)* and NaCl *(yellow)* transepithelial transport; arrow widths suggest relative transport magnitudes. Isotonic fluid is considered to have the same osmolality as blood plasma. Flow entering the AVR is assumed to arise from a descending vas rectum that is in, or near, a vascular bundle. Outflow from the collecting duct enters the inner medullary collecting duct. Tubular fluid flow direction is indicated by *blue arrows;* increasing osmolality is indicated by darkening shades of blue. *Thick black lines* indicate that a tubule is impermeable to water; *thin lines* indicate high permeability to water. *AVR,* Ascending vasa recta; *CD,* collecting duct; *IS,* inner stripe; *OS,* outer stripe.

from the smooth muscle contraction that gives rise to the peristaltic wave.[479] In the postwave decompression, the matrix exerts an elastic force that promotes water absorption from thin descending limbs and collecting ducts and thereby increases tubular fluid osmolality. Water absorption from the descending limbs would raise tubular fluid NaCl concentration and thus promote vigorous NaCl absorption from the loop bends and early ascending limbs. However, if, as is apparently the case in rats, the lower 60% of inner medullary descending limbs are water impermeable,[20] water is unlikely to be absorbed from descending limbs in the deep portion of the inner medulla, where the highest osmolalities are achieved. Alternatively, Knepper et al[84] proposed that the periodic compression of the papilla and the effects of that compression on the hyaluronan matrix could explain the osmolality gradient along the inner medulla. The proposed mechanism is consistent with the nodal compartments found by Pannabecker and Dantzler.[56,60] These compartments, which are likely rich in hyaluronan, are in contact with collecting ducts, thin ascending limbs, and ascending vasa recta and are thus well configured to be sites of transduction (i.e., sites where the mechanical energy of peristalsis is harnessed to generate an ascending flow that is dilute relative to average local osmolality).

DETERMINANTS OF CONCENTRATING ABILITY

In addition to the active transport of NaCl from the thick ascending limbs and the water permeability of the collecting ducts, other factors play a significant role in determining the osmolality of the final urine. On the nephron level, one major determinant is the delivery rate of NaCl and water to the loop of Henle, which sets an upper limit on the amount of NaCl actively reabsorbed by the thick ascending limb to drive the outer medullary concentrating mechanism. Another key determinant is the volume of tubular fluid delivered to the medullary collecting duct, which has an underappreciated effect on the concentrating process. Too much fluid delivery saturates water reabsorption processes along the medullary collecting ducts, leading to interstitial dilution due to rapid osmotic water transport. In contrast, too little fluid delivery results in sustained osmotic equilibration across the collecting duct epithelium.[141,145,146]

Demographic factors such as sex and age also influence urine concentration. Urine osmolality is generally higher in men than women, which suggests sex differences in thirst and AVP actions.[480,481] The kidney's ability to maximally concentrate urine also declines with age.[482] Older healthy men have higher AVP levels but a lower urine osmolality,[483] suggesting decreased sensitivity to AVP's antidiuretic effect, which may be attributed to structural differences in the aging kidney. Studies in aged rats have also suggested a decrease in many key transport proteins in the urine-concentrating mechanism, including aquaporins, urea transporters, and the V_2R, which would likely limit the kidney's response to water restriction.[482]

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